Dark-portal interactions of high-energy astrophysical neutrinos

Working notes

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1 Summary

What the code computes, and where it stands. We compute the diffuse flux of high-energy astrophysical neutrinos at IceCube in the dark-portal model of the theory notes (`notes/main.pdf`, "Notes on the theoretical framework for generalized new neutrino interactions"). A dark vector boson Z_D couples to active neutrinos through their mixing with a sterile state and to matter through Z–Z_D mass mixing. Two effects follow: neutrinos scatter resonantly on the relic (cosmic) neutrino background, which absorbs and regenerates them, and neutrons source a long-range coherent potential that changes their flavor in flight. The solver integrates both across cosmological distances, the Milky Way and the Earth; it is validated at four benchmark parameter points and runs in 0.71 s at its production defaults on 8 cores (1.04 s on one). The statistical analysis fits the predicted flavor composition to a projected 2050 measurement; it carries no explicit systematic uncertainties yet. The next stage, agreed on 2026-10-12, adds IceCube event rates, systematic uncertainties and coverage tests (Sec. 7). On 2026-10-12 we corrected the relic-neutrino temperature, every neutron number that sources the potential and the length of the Earth chords, and we now compute the Earth's own potential as the full Yukawa integral over its neutrons (Appendix A). The same day we removed the chord-sampling floor of the Earth transit, which had left $\sim 10^{-2}$ of noise in the predicted flavor fractions (Appendix A.7). Late that day we found three errors in the solver itself, present since its first version, and corrected them: the cosmological cascade created neutrinos without bound, the resonant absorption did not depend on the size of the couplings, and the propagator of the Milky Way and Earth transits was not unitary (Appendices A.8–A.10); on 2026-10-13 we made every absorption feed both of its secondaries, a neutrino and an antineutrino, where only one had been kept (Appendix A.11). Also on 2026-10-13 we redesigned the transit from the cosmological arrival to the Earth (Appendix A.12): adiabatic through the Galaxy, at its Yukawa-smoothed density, and coherent through the Sun's potential and the Earth's exterior, with eigenvectors that stay accurate at potentials 10^{40} times the vacuum term; the cosmological potential no longer stops growing below 3×10^{-13} eV. Every result of the statistical analysis, and every effect size quoted for the earlier corrections, was obtained before these fixes. Also on 2026-10-13 the project widened to secret neutrino interactions in general (Sec. 6): a scalar mediator with the full cross sections of arXiv:2107.13568, validated against that paper and run through the flavor-triangle analysis.

Table 1 lists the state of each component. Every number in these notes comes from a run of

the code, identified by the commit that contains it or by the test that checks it; anything not yet traced is marked [**VERIFY**]. Corrections to the code and to earlier statements are kept in Appendix A, because they are the reason the remaining numbers can be trusted. The agreed order of work is also kept, in shorter form, in `docs/dev/WORK_PLAN.md`.

Table 1: **State of each component, as of 2026-10-12.** "Validated" means checked at the four benchmark points of Sec. 4 with the acceptance rules stated there.

Component	State	Section
Flux solver	Validated; cascade, absorption normalization and propagator corrected 2026-10-12; 0.71 s on 8 cores	3, 4
Thermal absorption rates	Semi-analytic kernel; relic temperature corrected 2026-10-12	3.2, A.1
Earth transit	Phase-exact stepping with a step-count certificate; 25 steps; the Earth's potential as its Yukawa integral	3.5, A.6
Flavor-ratio analysis	Results predate the solver corrections and are void; being rebuilt	5
IceCube event rates	Planned; data sets and folding method chosen	7.1
Systematics and coverage	Planned; list and procedure agreed	7.2, 7.3
Neutron densities	All corrected 2026-10-12	2.4, A
Earth-chord sampling	Floor removed by the phase-aware zenith average (2026-10-12)	3.5, A.7
Transit to the Earth	Redesigned 2026-10-13: adiabatic Galaxy, coherent Sun and Earth exterior; checked against brute force	A.12
Secret interactions	Scalar mediator, Majorana neutrinos, full cross sections (2026-10-13); validated; flavor-triangle planes. Dirac and vector next	6

2 The model and its implementation

2.1 Where the physics is defined

The theory notes define the framework; we cite their equations as "Eq. (Tn)". The neutrino ensemble is a 3×3 density matrix $\rho(E_0, z)$, where E_0 is the energy observed at Earth and z the redshift. It obeys the quantum kinetic equation (T3). The Hamiltonian (T4) drives its coherent part; the collision operator (T6) is its incoherent part. The coherent part contains the standard matter potential (T5) and the long-range dark-portal potential (T9) for neutrinos, (T10) for antineutrinos. The incoherent part contains the attenuation rate (T7) and the regeneration kernel (T8), evaluated with a Breit–Wigner resonance of width (T11) and coupling (T12). The sources follow a power law with a free flavor composition (T18), distributed in redshift as the star-formation rate (T19, T20). The thermal (Debye) mass that screens the mediator is (T23).

Parameters.—The dark sector is set by the mediator mass m_{Z_D}, the dark gauge coupling g_D, the Z–Z_D mass mixing ϵ', and the active–sterile mixing magnitudes $u_\alpha \equiv |U_{\alpha 4}|$ ($\alpha = e, \mu, \tau$). Resonant absorption on a relic neutrino of mass m_i peaks near $E_{\rm res} = m_{Z_D}^2/(2m_i)$, so a mediator of mass $\sim$ 1–10 MeV places the absorption features inside the IceCube energy range. Table 2 lists every parameter with its default value and configuration key. The defaults live in one place, `src/dp_config.py` (`DEFAULT_CONFIG`); the analysis configurations carry their own copies of the physical constants.

Table 2: **Parameters and their defaults.** Keys refer to the solver's configuration sections; the analysis configurations use the same names under `physics_constants`. Masses and energies in eV unless stated.

Symbol	Meaning	Default	Key
m_{Z_D}	Dark mediator mass	2×10^6	`dark_portal.m_Zd_eV`
g_D	Dark gauge coupling	10^{-4}	`dark_portal.g_D`
ϵ'	Z–Z_D mass mixing	0.1	`dark_portal.epsilon_prime`
u_e, u_μ, u_τ	Active–sterile mixing	$0.5, 0.5, 0$	`dark_portal.u_*`
p	Source spectral index	2.5	`source.p_index`
$f_{e,S}, f_{\mu,S}, f_{\tau,S}$	Source flavor composition	$1/3, 2/3, 0$	`source.f_*_S`
$z_{\max}$	Largest source redshift	4	`source.z_max`
m_{lightest}	Lightest neutrino mass	0.01	`cosmology.m_lightest_eV`
$T_{\nu,0}$	Relic-neutrino temperature today	1.676389×10^{-4}	`cosmology.T_nu0_eV`
$n_{n,0}$	Cosmological neutron density	3.091×10^{-8} cm^{-3}	`cosmology.n_n0_cm3`
$\theta_{12}, \theta_{13}, \theta_{23}$	Mixing angles	$33.41^\circ, 8.58^\circ, 49.1^\circ$	`standard_mixing.th*_deg`
δ_{CP}	Dirac phase	132°	`standard_mixing.d_cp_deg`
$\Delta m^2_{21}, \Delta m^2_{31}$	Mass splittings [eV^2]	$7.41 \times 10^{-5}, 2.511 \times 10^{-3}$	`standard_mixing.dm*_sq`

2.2 From the equations to the code

The solver (`src/`, entry point `run_solver` in `src/dark_portal_solver.py`) divides the propagation into three phases. Each phase hands its result to the next, and neutrinos and antineutrinos run separately before their fluxes are combined.

Cosmological phase.—From $z_{\max} = 4$ to $z = 0$ the flux is carried as populations in the instantaneous eigenbasis of the Hamiltonian (T4), one per mass state, with no coherences between them. At each step the populations decay at the thermal absorption rate (T7) and receive the regenerated flux of the cascade kernel (T8) and the injected source flux (T18). We project them onto flavor with the mixing probabilities $|U_{\alpha i}(E, z)|^2$ of the full Hamiltonian. The cascade kernel is separable, which reduces the regeneration integral to a suffix sum that costs $O(N_E)$ per step instead of $O(N_E^2)$ (`step_cosmological_propagation` in `src/dp_cosmo.py`).

Transit to the Earth (since 2026-10-13).—Through the Galaxy the populations are carried adiabatically: they become the same populations of the local eigenstates at the Sun, where the potential is the Yukawa integral of the Galaxy's neutrons over the mediator's range. Through the Sun's potential and the Earth's exterior the density matrix is integrated coherently in the instantaneous eigenbasis, along eight directions to the Sun and one radial path for the Earth (`src/dp_transit.py`, `transit_to_earth` in `src/dp_spatial.py`; Appendix A.12). Until 2026-10-13 a Milky Way phase evolved the density matrix from 100 kpc along six directions in 25 steps each.

Earth phase.—The density matrix is then evolved along $n_\oplus$ chords through the Earth (PREM densities, optional standard matter potential) to a detector 1.5 km deep, starting at twice the Earth's radius below $m_{\text{eff}} = 2 \times 10^{-5}$ eV, where the exterior matters. The chords are evenly spaced in the cosine of the zenith angle over the whole sky, and the zenith average integrates their dark-portal phase (Sec. 3.5). The flavor fluxes at the detector are the diagonal of the averaged density matrix.

Antineutrinos.—The antineutrino run uses U^*_{PMNS} and the transformed potential $-V^*$ (T10). With the default `is_antineutrino = "sum"` the two runs are added, weighted by `nubar_to_nu_ratio` ($= 1$).

2.3 Potential textures

The flavor structure of both coherent potentials can be replaced through the configuration: `dark_portal.U_LRI_struct` for the dark-portal potential (default $u_\alpha u^*_\beta$, as in Eq. (T9)) and `standard_mixing.V_SM_struct` for the standard matter potential (default diag$(1, 0, 0)$, as in

Eq. (T5)). Each is a 3×3 nested list whose entries are real numbers or [Re, Im] pairs. The regression tests `test_structure_override_*` and `test_parse_structure_matrix_3x3_formats_and_validation` check the parsing and the antineutrino convention. In addition, we ran rank-zero, full-rank complex and indefinite textures through both spatial phases and the cosmological mixing table (2026-07-29); any fixed Hermitian texture works.

Limits of the override.—The texture must be Hermitian to 10^{-12} relative precision; the parser rejects anything else, because the propagator reads only one triangle of the Hamiltonian. It must be constant within a run, since the cosmological mixing table assumes a Hamiltonian of the form $M + \sigma S$ with one fixed matrix S. Overrides are given in the neutrino convention; the solver applies the antineutrino transformation itself. The override changes only the coherent potential: the Debye mass and the absorption rates remain set by u_e, u_μ, u_τ. Its overall scale is degenerate with $g_D\epsilon'$.

2.4 Physical inputs

Sources.—A diffuse population following the star-formation history up to $z = 4$, with spectrum $E^{-2.5}$, flavor ratios 1/3 : 2/3 : 0 (pion decay) and equal numbers of neutrinos and antineutrinos. At each redshift the injected flavors split incoherently over the eigenstates of the Hamiltonian there; by default the sources sit in the cosmic mean neutron density, and `source.host_galaxy_scale` places them inside a Milky-Way-like host instead (Sec. 5).

Relic neutrinos.—The absorption rates integrate the Fermi–Dirac distribution of the relic neutrinos at temperature $T_{\nu,0}(1 + z)$. Therefore, the relic density in our rates follows from $T_{\nu,0}$ and scales as $T_{\nu,0}^3$. The nominal 56 cm^{-3} per state enters only as a ratio that carries flavor asymmetries. The temperature is $T_{\nu,0} = (4/11)^{1/3}\, T_{\rm CMB} = 1.9454$ K $= 1.676389 \times 10^{-4}$ eV for $T_{\rm CMB} = 2.7255$ K. Until 2026-10-12 the code used 1.95×10^{-4} eV (Appendix A.1).

Neutron densities.—The dark-portal potential is proportional to the neutron number density along the path, because the Z-like couplings of protons and electrons cancel in neutral matter. Cosmologically, the neutrons sit almost entirely in helium-4, $Y_p/2$ per baryon, so $n_{n,0} = n_b Y_p/2 = 3.091 \times 10^{-8}$ cm^{-3} for $n_b = 2.515 \times 10^{-7}$ cm^{-3} (PDG 2024) and $Y_p = 0.2458$ (LBT Y_p project, 2026). In the Galaxy we convert the stellar mass density at 0.20 neutrons per nucleon, the electron density of the hot halo at 0.15 neutrons per electron, and the cold HI and H_2 discs of McMillan (2017) at 0.142 neutrons per nucleon (Appendices A.2 and A.4). The Sun, the Moon and the Earth also act as point sources, with 1.929×10^{56}, 2.232×10^{49} and 1.849×10^{51} neutrons. In the Earth we use $1 - Y_e$ neutrons per nucleon, with electron fractions $Y_e = 0.4656$ in the core and 0.4957 in the mantle and crust, and the composition of water for PREM's top layer (Appendix A.3).

The Earth's own potential.—The potential at a point is a Yukawa integral over the neutrons around it, with range $1/m$, $m = (m_{Z_D}^2 + m_D^2)^{1/2}$. Distant bodies need only its far limit, a point source; the cosmic mean density, uniform, gives n/m^2, cut at the Hubble radius (Appendix A.12). A local (contact) density holds only where $1/m$ is short against every length over which the density varies: for the Galaxy, below $\sim 10^{-25}$ eV, we integrate its model over the range (Appendix A.12). Inside the Earth the range can match the size of the body, so we evaluate the integral itself. For a spherical profile it is one-dimensional,

$$n_{\rm eff}(r) = m^2 \left[\frac{e^{-mr}}{r} \int_0^r r'^2 n(r') \, \frac{\sinh mr'}{mr'} \, dr' + \frac{\sinh mr}{mr} \int_r^R r' n(r') \, e^{-mr'} dr' \right], \qquad (1)$$

and the potential is the contact expression evaluated with $n_{\rm eff}$ in place of n. This is the form used for the long-range potential in the Sun in Magnus and in the Earth in NuOscProbExact (notebook 20). Appendix A.6 gives our evaluation and its tests.

3 Numerical method

3.1 Resolution and defaults

Table 3 compares the solver's production defaults with the settings of the statistical analysis. The defaults follow the convergence study of 2026-07-28 (`tools/convergence_study.py`, results in `output/convergence/`). We remeasured the analysis settings on 2026-10-12 with the corrected solver (Appendices A.8 to A.10): the errors of the study, measured on the runaway cascade, were up to two orders of magnitude too large. The analysis uses coarser energy and redshift grids than the defaults.

Table 3: **Resolution settings.** The convergence column states what each setting is known to achieve.

Setting	Solver default	Analysis	Evidence
Energy bins N_E (10 TeV–30 PeV)	1000	400	$\leq 1.5 \times 10^{-5}$ on flavor fractions at 400 (inband), 2×10^{-4} at $m_{Z_D} = 10^{-2}$ eV
Redshift steps N_z ($z \leq 4$)	1000	200	$\leq 7 \times 10^{-6}$ at 200
Directions to the Sun	8	8	χ^2 within 0.03 of 32 directions (Appendix A.12)
Transit table spacing	0.002	0.002	$\sim 10^{-5}$ against brute force; χ^2 within 0.03 of 0.001
Earth chords $n_\oplus$	12	12	12 agree with 50 to $\leq 1.4 \times 10^{-7}$ (Appendix A.7)
Earth steps N_{Earth}	25	25	Certified, Sec. 3.5

The Earth-chord floor.—Until 2026-10-12, changing the number of chords moved the energy-averaged flavor ratios pseudo-randomly at the 10^{-2} level instead of converging. We had attributed this to energy fringes finer than the grid; it was sampling noise of the dark-portal phase, which does not depend on energy, in the equal-weight chord mean. The phase-aware zenith average removes it (Appendix A.7). With the chord floor gone and the solver corrected, the largest numerical error left at the analysis settings is the energy grid, up to 2×10^{-4} on the fractions at $m_{Z_D} = 10^{-2}$ eV, where fine fringes in energy need ~ 1000 bins; at the benchmark points it is below 1.5×10^{-5}.

3.2 Thermal absorption rates

The rates (T7) require a momentum integral over the relic distribution of a Breit–Wigner resonance whose width is far smaller than any practical grid. We integrate it semi-analytically: the momentum at which the resonance enters the angular range has a closed form, the narrow zone around it is integrated with exact antiderivatives, and the smooth remainder with Gauss–Legendre quadrature (`src/dp_thermal.py`). Against a dense reference integration, the significance-weighted error is 2.5×10^{-7}; the midpoint rule used before July 2026 erred by 6–9% at on-resonance cells (round 7, commits `53d74f2`–`ca816a4`).

Coupling cache.—Within a fit at fixed m_{Z_D} the rate grid depends on the couplings only through the total coupling s, as $U(s) \approx I_A + I_B/s$ to $\sim 7 \times 10^{-8}$ over brackets of up to two decades. The cache fits this form from two exact grids and uses it only after checking it against a third (tolerance 3×10^{-7}). Until 2026-07-29 its minimum bracket width (0.3 decades) exceeded the coupling steps of the minimizer, so it never fired in real scans. We replaced that width with a numerical floor. The cache then served 83% of the evaluations of minimizer-like sequences at three mediator masses, with a worst error of 6.5×10^{-9} against fresh exact grids (commit `5e7877c`).

3.3 Cosmological mixing

The eigenvectors of $M/(2E(1+z))+A(z)S$ depend on E and z only through $\sigma = 2E(1+z)A(z)$, because eigenvectors are unchanged by an overall factor. Therefore, we tabulate the mixing probabilities on 4096 nodes in $\log\sigma$ and interpolate, instead of computing $\sim 10^6$ eigenvector sets per run. Our table agrees with direct computation to 2.7×10^{-14} at a stress point with a level crossing in range (`README.md`, Sec. 12). Since 2026-10-13 the eigenvectors come from a Jacobi iteration in the texture's eigenbasis, accurate at any σ; the closed form used before lost the vacuum splitting of the rank-one texture's two degenerate eigenvalues once σ exceeded M by $\sim 10^{13}$ (Appendix A.12). The table now stops where the eigenbasis has become the texture's own to 10^{-9}.

3.4 Coherent propagation

The evolution operator of each step, $\exp(-iH\Delta x)$ for a 3×3 Hermitian H, is the spectral sum $\sum_k e^{-i\lambda_k \Delta x} v_k v_k^\dagger$ over an eigenbasis built to stay orthonormal to rounding at any eigenvalue gap (`src/dp_propagator.py`). Closed-form roots of the traceless part order the eigenvalues; the isolated one's eigenvector is a column of the product of the matrices shifted by the other two; the remaining pair comes from a 2×2 block. The operator is therefore unitary to 10^{-15} at any phase (Appendix A.10). A pair whose splitting H does not resolve in double precision evolves as degenerate; under a dominant potential this drops the vacuum splitting inside the potential's degenerate subspace, which changes flavor populations by at most 5×10^{-6} across the Earth's diameter at 10 TeV. When the potential is so large that double precision cannot follow the oscillation phase, the fluxes carry only the decoherence-averaged information. This happens roughly where $m_{Z_D}^2 + m_D^2 \lesssim 3 \times 10^6\, g_D \epsilon'$ eV^2, i.e., below $m_{Z_D} \approx 5$ eV for the default couplings ($g_D = 10^{-4}$, $\epsilon' = 0.1$); the analysis warns when its scan window reaches this region.

3.5 Earth steps

In each Earth step we use the exact average of the PREM density over the step, integrated layer by layer with the layer crossings in closed form. When the mediator's range approaches the thickness of a layer, the density of Eq. (1) relaxes across each jump over a length $1/m$; we integrate these boundary layers on subintervals graded toward them. The accumulated phase then equals the integral of the potential along the chord for any number of steps. Fluxes computed with 8 and with 1000 steps agree to $\leq 4.5 \times 10^{-9}$, and their energy-averaged flavor ratios to $\leq 6.5 \times 10^{-10}$, at the three benchmark points of Sec. 4 that are not in the phase-noise regime (commit `e1b6660`). Since the scheme converges to the exact path-ordered result as the steps shrink, this independence of the step count shows that the result is converged. The earlier left-endpoint scheme accumulated a phase error of thousands of radians, which moved the fluxes by up to 10^{-2} whenever the step count changed. The default is 25 steps since 2026-07-29 (commit `95fef9f`, Table 4).

Zenith average.—The chords are combined by integrating each block of their density matrices against its dark-portal phase over $\cos\theta_z$ (Appendix A.7), not by an equal-weight mean.

The whole sky (2026-10-13).—A detector sees neutrinos from every direction. Since 2026-10-13 the average runs over the whole sky with equal weight per $\cos\theta_z$ (the decided default; up-going only remains an option): down-going neutrinos cross only the ice above the detector, 1.5 km straight down to ~ 140 km at the horizon, against up to 12,740 km through the Earth. The down-going chords mirror the up-going ones, and the phase-aware average reproduces the exact integral over the sky to 10^{-5} (a test against 400,001 directions). How much the sky dilutes the Earth's effect depends on the potential: by half while the ice adds little phase (a dark-portal phase through the core below ~ 50 rad), less as the near-horizon paths start to mix flavors, not at all once the ice mixes them from above too. At the MeV benchmarks the phase is 6×10^5 rad

through the core, 680 rad at the horizon and 7 rad straight down, so near-vertical down-going neutrinos keep part of the flavor structure of the flux arriving at the Earth: f_e and f_μ, equal from below, differ by 10^{-3} at "inband" and 3×10^{-3} at "strong". The detector's own acceptance per direction will replace the equal weights in the event-rate analysis (Sec. 7.1).

3.6 Parallelism and cost

Earth chords are independent; they run in a persistent pool of processes started with `spawn`, the start method the analysis requires. The transit to the Earth runs on threads, over energy bins, and only in the bins that a source actually turns: ~ 6 s per evaluation where the Sun matters (400 bins, 8 directions), nothing elsewhere. The cosmological phase and the thermal grids use threads (numba, TBB layer). On the development laptop (Intel i5-1334U, 8 cores), a solver run at the defaults took 0.32 s on 2026-07-29 and takes 0.71 s now (Sec. 8). One likelihood evaluation at the analysis settings (400×200, 6 paths, 12 chords) takes 0.35 s on one core (2026-10-12). Below $m_{\rm eff} \approx 2 \times 10^{-5}$ eV the Earth's Yukawa columns depend on the couplings through the Debye mass and are recomputed for every new point, adding ~ 0.3 s; they are cached per $m_{\rm eff}$, so the antineutrino pass reuses those of the neutrino pass. Before the step count fell from 100 to 25, the Earth phase took 66% of a single-core evaluation and the Milky Way phase (replaced on 2026-10-13) 18%.

4 Validation

4.1 The four benchmark points

Every change that can affect the physics is checked against the full solver at production settings ($N_E = N_z = 1000$, 6 paths, 12 chords, 4 cores) at four parameter points, with the reference and the modified code in separate processes:

Point	m_{Z_D} [eV]	g_D	ϵ'
below	3×10^5	10^{-4}	0.1
inband	2×10^6	10^{-4}	0.1
strong	2×10^7	10^{-2}	0.1
cosmoX2	40	10^{-4}	0.1

At "below" all three absorption resonances lie under the energy range; at "inband" they lie inside it; "strong" has a hundred times the coupling. At "cosmoX2" the mediator is light, so the potential dominates the Earth transit. Until 2026-10-12 a change in the last floating-point digit moved its pointwise fluxes by $\sim 10^{-3}$, which we attributed to the large phase; it was the non-unitary propagator (Appendix A.10). It now moves them by 1.8×10^{-12}.

Acceptance rules.—We compare two quantities: the largest relative flux difference over all energies and flavors, and the energy-averaged flavor ratios. A change that should not alter the arithmetic must reproduce "below", "inband" and "strong" to the floor at which two runs of identical code agree, ~ 2–7×10^{-12}. The "strong" point has an additional common-mode floor of $\sim 7.5 \times 10^{-8}$ under any reordering of the last floating-point digit, with its flavor ratios at $\lesssim 10^{-12}$. Until 2026-10-12 we judged "cosmoX2" on the energy-averaged flavor ratios and the energies of the absorption dips only; it now meets the same rules as the other points. Any recompilation of the spatial kernels changes the results at the last digit: two textually identical kernels compiled separately differed by 3.7×10^{-16}.

Absolute checks.—Comparing a modified code with its predecessor cannot reveal an error that both share; three such errors survived from the first version of the solver until 2026-10-12 (Appendices A.8–A.10). We now also check every change against conservation: the energy and number of neutrinos reaching the Earth must not exceed those of the same run without interactions, the spatial phase must conserve every energy bin's total, and vanishing sterile mixing

must reproduce standard propagation. At production settings the energy (number) reaching the Earth is 0.99910 (0.99827) of the no-interaction value at "below", 0.94135 (1.09231) at "inband", 0.99995 (1.00074) at "strong" and 0.99944 (0.99901) at "cosmoX2"; the energy-averaged fractions are $(0.32903, 0.32896, 0.34201)$, $(0.33241, 0.33138, 0.33621)$, $(0.33050, 0.32751, 0.34199)$ and $(0.32899, 0.32899, 0.34201)$ over the whole sky ($f_e = f_\mu$ from below: Sec. 3.5), and change by $\leq 1.2 \times 10^{-7}$ between 1000 and 2000 redshift steps (2026-10-13, Appendix A.11). The number can exceed the no-interaction value: each absorption turns one particle into two.

4.2 Record of validated changes

Table 4: **Changes checked at the benchmark points.** Differences are the largest relative flux difference and the largest change of an energy-averaged flavor fraction.

Date, commit	Change	Result
2026-07-29 `e1b6660`	Phase-exact Earth stepping	Step-count independent to 4.5×10^{-9} (Sec. 3.5)
2026-07-29 `0d6392c`	Solver split into modules	Identical at inband, strong, cosmoX2; below at the 7×10^{-12} floor
2026-07-29 `95fef9f`	Earth steps $100 \to 25$	below 1.0×10^{-9} / 1.2×10^{-10}; inband 2.1×10^{-11} / 2.3×10^{-12}; strong 6.2×10^{-12} / 8.3×10^{-13}; cosmoX2 ratios 2.9×10^{-3}, dips unchanged
2026-10-12 `5ecf10e`	Relic temperature corrected	A physics change; effects in Appendix A.1
2026-10-12 `0db8111`	Cosmological and Galactic neutron densities	No change at the four points beyond their floors; effects at light mediators in Appendix A.2
2026-10-12 `9c025e6`	Neutron fractions along Earth chords	Flavor fractions move by up to 0.026, as much as adding one chord (Appendix A.3)
2026-10-12 `2dec771`	Point sources and cold disc gas	No change at the four points beyond their floors; effects at light mediators in Appendix A.4
2026-10-12 `5866227`	Exact length of the Earth chords	Flavor fractions move by 9×10^{-4} (below) and $\leq 5 \times 10^{-5}$ (inband, strong); dips unchanged (Appendix A.5)
2026-10-12 `70ef709`	The Earth's potential as its Yukawa integral	Bit-identical at the four points; effects where the range reaches the Earth's scale in Appendix A.6
2026-10-12 `8f1910b`	Phase-aware zenith average of the Earth chords	Chord-sampling floor removed; flavor fractions move by up to 0.029 (Appendix A.7)
2026-10-12 `7dc95e9`	Cosmological cascade conserves number	A correction: the energy at the Earth was up to 10^{67} times the injected one (Appendix A.8)
2026-10-12 `944360d`	Resonant absorption scales with the couplings	A correction: energy at the Earth $3 \times 10^{-3} \to$ 0.999 of the no-interaction value at "below" (Appendix A.9)
2026-10-12 `2066f3b`	Exactly unitary propagator	below, inband, strong unchanged to five digits; "cosmoX2" fractions move by 3.5×10^{-4} and become reproducible (Appendix A.10)
2026-10-13	Both secondaries of every absorption	inband fractions move by 1.8×10^{-3}; energy at the Earth $0.841 \to 0.941$ (inband), $0.982 \to 0.99995$ (strong) (Appendix A.11)
2026-10-13	The whole sky at the detector	$f_e - f_\mu$ from 0 to 10^{-3} (inband), 3×10^{-3} (strong); energies unchanged (Sec. 3.5)
2026-10-13	The transit to the Earth; the cosmological potential at ultralight masses	below, inband, strong to 3×10^{-12}; cosmoX2 1.4×10^{-3} pointwise, 1.8×10^{-8} on the fractions, the old method's own error (Appendix A.12)

5 Statistical analysis: current state

5.1 Likelihood and parameters

The analysis (`analysis/`, which replaced FrequentistAnalysis on 2026-10-12; `docs/dev/ANALYSIS_DESIGN.md`) predicts the flavor composition at Earth, $f_\alpha = \int \Phi_\alpha \, dE / \int \Phi \, dE$ over 10 TeV–30 PeV, and compares it with a projected measurement: an asymmetric Gaussian fitted to the expected 2050 sensitivity of IceCube with IceCube-Gen2, KM3NeT, Baikal-GVD, P-ONE and HUNT, centred on full pion decay (surface `pion_2050_case_5` in `analysis/data/flavor_surfaces.json`, one of 35, whose $\ln L$ we reproduce bit for bit). We use $\chi^2 = -2\ln(L/L_{\max})$, zero at the centre. Detector uncertainties enter only through the width of that projection. In the two analyses ported from FrequentistAnalysis four parameters float, m_{Z_D}, g_D, u_e and u_μ (three at $m_{Z_D} = 10^{-30}$ eV); everything else is fixed, including the source composition, the spectral index and the lightest neutrino mass. The mixing parameters are fixed at the NuFIT 6.0 best fit for normal ordering with Super-Kamiokande atmospheric data ($\sin^2\theta_{12} = 0.308$, $\sin^2\theta_{13} = 0.02215$, $\sin^2\theta_{23} = 0.470$, $\delta_{\mathrm{CP}} = 212°$), which reproduces the projection's centre to 10^{-6}. FrequentistAnalysis fixed them at the solver's defaults ($\delta_{\mathrm{CP}} = 132°$, $\sin^2\theta_{23} = 0.571$), which put standard physics 0.0034 away from the centre in f_μ, 0.7σ of the projection; only new physics could absorb that offset.

Limitations we found.—The flavor fractions are plain integrals of the flux; a detector would weight them by its effective area and cross sections, which rise with energy. For an $E^{-2.5}$ spectrum the plain integral is dominated by energies near 10 TeV. The event-rate calculation of Sec. 7.1 removes this approximation. The priors that FrequentistAnalysis' configurations defined for the mixing parameters were never applied (Appendix A.13); the new package applies Gaussian and half-Gaussian priors, also on derived quantities such as the sum of the neutrino masses.

5.2 Scans and limits

Each fit is a bounded quasi-Newton minimization (L-BFGS-B) from several seeded starts, polished by Powell's derivative-free method; parameters spanning decades are fitted in $\log_{10}$. A profile fixes the parameter of interest on a grid and minimizes over the others; a plane does the same for two parameters. The grid points are fitted in parallel, in chains walked outward from the global fit, each point starting from its predecessor's solution and from the global fit. Repair passes then refit points that their neighbours' solutions beat, points that stand out above their neighbours, and the edges of the grid. Every fit is checkpointed, so an interrupted scan resumes. We read intervals off $\Delta\chi^2$, measured from the lowest χ^2 found anywhere, at 1, 2.706 and 3.841 for one parameter (68.27%, 90%, 95%, two-sided) and at 2.296, 4.605 and 5.991 for two; FrequentistAnalysis labelled 4.61 as 2σ.

Exclusion planes (decided 2026-10-13).—We read an exclusion plane per mediator mass: in each column of fixed m_{Z_D}, $\Delta\chi^2$ is measured from that column's own minimum, and the couplings above $\Delta\chi^2 = 3.84$ are excluded at 95% (a raster scan, one parameter per column). The two readings differ most where the effect saturates: on the plane below, the Earth's potential alone gives $\Delta\chi^2 \approx 4.5$ over most light masses, above 3.84 but below the 5.99 of a joint two-parameter region. A column whose own minimum lies above the best fit by more than 3.84 is flagged: its couplings never reached the region without an effect, so its range must be extended before a limit is read. Coverage tests still have to confirm the threshold; for a coupling bounded at zero the asymptotic one-sided value is 2.71.

Planes by regime (decided 2026-10-13).—The observables depend on g_D and the sterile mixing only through products, so a free mixing would always slide to zero; the planes scan the products, and their limits hold for any mixing. MeV planes set $\epsilon' = 0$, which switches off the coherent potential and leaves the resonant absorption, a function of $g_D|U_{\alpha 4}|^2$ alone.

Potential planes run from ultralight to GeV mediators in $g_D\epsilon'|U_{\alpha4}|^2$, to which the potential is proportional; with $\epsilon' = 0.1$ the absorption coupling $\Sigma G \approx g_D|U_{\alpha4}|^2\,\epsilon' g/c_W$ depends on the same product. At MeV masses the Earth's own potential is far from negligible: $\sim 10^{-8}$ eV at $g_D\epsilon' = 10^{-5}$ and 2 MeV, against a vacuum term of $\sim 10^{-16}$ eV. Toward lighter mediators the potential at the detector comes from the Earth's exterior, then the Sun (10^{-26} to 10^{-18} eV), then the Galaxy smoothed over the mediator's range (Appendix A.12). Each regime is run for the three benchmark textures (T13), (T16), (T17). A test checks that splitting a product differently leaves the fractions unchanged (10^{-12} with $\epsilon' = 0$; 10^{-6} for the potential).

The potential planes (2026-10-13, commit `52f36fc`, Tycho).—With the transit of Appendix A.12 the three potential planes (40×55 points each, ~ 3 minutes on 16 cores) follow the source that dominates the potential at the detector. For the $e\mu$ texture the lowest excluded coupling product falls from 3×10^{-48} to 2×10^{-55} between 10^{-24} and 10^{-18} eV (the Sun) and from 3×10^{-47} to 5×10^{-51} around 10^{-16}–10^{-14} eV (the Earth's exterior), and rises from 10^{-60} to 2×10^{-57} at 10^{-30} eV (the Galaxy, no longer overstated); above 10^{-8} eV nothing changes. The $\mu\tau$ texture moves the same way (by up to 15 decades at 10^{-18} eV). For the $e\tau$ texture the potential-dominated composition happens to lie close to the standard one: the excluded region is the transition itself, where the Galactic or solar potential crosses the vacuum term, and the region above $g_D\epsilon'|U|^2/m_{Z_D}^2 \approx 10^4$ eV^{-2}. Every scattered point of the old planes is gone.

A band set by the sources' environment.—All three planes have a narrow band along $g_D\epsilon'|U|^2/m_{Z_D}^2 = 10^4$ eV^{-2}, one grid row wide, where $\Delta\chi^2$ dips (1.3 for $e\mu$, inside the excluded region). There the cosmological potential at the sources ($z \sim 1$) equals the vacuum term, so the injected flavors project halfway between the vacuum states and the potential's, and the composition at the Earth passes close to the standard one. The band existed before above 3×10^{-13} eV; the mass floor of Appendix A.12 flattened it below, into the old planes' horizontal stripe at 10^{-21}. It assumes the sources sit in the cosmic mean density; for ranges shorter than ∼Mpc a source inside its host galaxy sees 10^6–10^7 times more neutrons, which would move the band. The source configuration therefore has a knob (decided 2026-10-13), `source.host_galaxy_scale`, off by default: with $s > 0$ the sources sit inside a Milky-Way-like host at a Sun-like position, its neutrons scaled by s, with the same Yukawa-smoothed profile as our own Galaxy (Appendix A.12) added to the cosmic mean and not diluting with $(1+z)^3$. Only the injection changes: the injected flavors split over the host's eigenstates, which leave the host adiabatically. With $s = 1$ the transition moves from $g_D\epsilon'|U|^2/m_{Z_D}^2 \approx 10^4$ to $\sim 10^{-3}$–1 eV^{-2} (a test: deep in it both settings give the same composition). The three potential planes with $s = 1$ (Tycho, commit `971e44b`): for $e\mu$ the lowest excluded coupling is unchanged in 37 of 40 mass columns and one grid row (a decade) lower in the other three (10^{-30}–10^{-26} eV); for $\mu\tau$ it moves by one row at most; for $e\tau$, whose high-coupling exclusion is this transition, the excluded region now starts at $g_D\epsilon'|U|^2/m_{Z_D}^2 \sim 10^{-2}$ instead of 10^3–10^4 eV^{-2}, which lowers the limit by ~ 7 decades between 10^{-25} and 10^{-17} eV. The $e\mu$ and $\mu\tau$ limits therefore do not depend on where the sources sit; the $e\tau$ limit below 10^{-17} eV does.

A first plane (2026-10-13; not for quoting; computed before the regeneration fix of Appendix A.11).—For the $e\mu$ texture at $u = 0.5$ and $\epsilon' = 10^{-4}$ (configuration `pion_2050_case_5_exclusion`, 45×41 points, commit `19f953d`), the 2050 projection excludes at 95% $g_D > 3.7 \times 10^{-9}$ at $m_{Z_D} = 0.1$ MeV, 3.3×10^{-7} at 1 MeV (with an allowed island at $10^{-5} < g_D < 0.013$, where the resonant absorption offsets the potential), 3.7×10^{-5} at 10 MeV and 3.7×10^{-3} at 100 MeV: the edge follows $g_D \propto m_{Z_D}^2$, where the Earth's potential reaches the vacuum term. Below ~ 30 keV the columns stay 4.3–4.7 above the best fit down to $g_D = 10^{-10}$, the bottom of the scan, so no limit can be read there yet. Three things keep this from being a result: $u = 0.5$ is far above laboratory bounds on the mixing, only up-going directions enter the average, which overstates the Earth's effect, and the light-mass columns need couplings scanned far lower.

Profiles of null data.—The projection is centred on standard physics, and with every coupling free the fit can always switch the new physics off ($u \to 0$, $g_D \to 0$, or a resonance outside the

energy range). The profiles of the ported analyses are therefore expected to stay at $\Delta\chi^2 \approx 0$, and an exclusion region needs fixed couplings: the configuration `pion_2050_case_5_exclusion.json` maps $\Delta\chi^2$ over (m_{Z_D}, g_D) for the $e\mu$ texture (T13) at $u = 0.5$.

Degeneracies.—Flavor fractions alone do not identify the parameters. We fitted Asimov data, the model's own prediction at $m_{Z_D} = 2$ MeV, $g_D = 3 \times 10^{-4}$, $u = 0.5$: with m_{Z_D} and g_D free the fit reached $\chi^2 = 7 \times 10^{-7}$ at $m_{Z_D} = 8 \times 10^{-14}$ eV (both the resonant absorption and the coherent potential pull f_e and f_μ together when $u_e = u_\mu$); with the mixing free as well it returned the injected mass to 5×10^{-6} with $g_D = 10^{-4}$ ($\chi^2 = 8 \times 10^{-4}$). Injection-recovery tests must therefore check that the fit reaches $\chi^2 \approx 0$, not that it returns the injected values.

What we have not shown.—Nothing yet checks that the intervals have their stated coverage. Near the boundaries of the parameter space (couplings at zero, the lightest mass at zero) the asymptotic thresholds can fail. For an upper limit on a coupling bounded at zero the one-sided values are 1.64 and 2.71, so limits read at the two-sided values are probably conservative. Section 7.3 describes the tests.

6 Secret neutrino interactions

On 2026-10-13 we widened the project beyond the dark portal (decided that day; the plan is in `docs/dev/WORK_PLAN.md`): the same machinery now tests secret neutrino interactions generally, as in arXiv:2001.04994 (Bustamante, Rosenström, Shalgar & Tamborra), with the full cross sections of arXiv:2107.13568 (Esteban, Pandey, Brdar & Beacom, Appendix A). The switch `secret_interactions.enabled` replaces the dark portal by this model; everything downstream (the analysis, the scanner) is unchanged. This section describes the scalar mediator with Majorana neutrinos (`src/dp_si.py`, commits `781ae43`, `771cd46`), then Dirac neutrinos and a vector mediator (`99a8ccc`, `d82231c`; Sec. 6.2), all implemented and validated on 2026-10-13, and the limits of the four models (Sec. 6.6).

6.1 Model and cross sections

A real scalar ϕ of mass M couples to the neutrinos as $-\frac{1}{2}\sum_{\alpha\beta} G_{\alpha\beta}\,\bar\nu_\alpha\nu_\beta\,\phi$, for any real symmetric flavor texture G (Mauricio, 2026-10-13: non-universal textures must be treated). For Majorana fields, $\nu_\alpha = \sum_i (U_{\alpha i}P_L + U^*_{\alpha i}P_R)\nu_i$, so $\bar\nu_\alpha\nu_\beta = \sum_{ij}\bar\nu_i[(U^TGU)^*_{ij}P_R + (U^TGU)_{ij}P_L]\nu_j$: the mass states couple through the complex symmetric matrix $\lambda = U^TGU$. The paper writes $g_{ij} = \sum U^*_{\alpha i}U_{\beta j}G_{\alpha\beta}$ and takes it real, which agrees only without CP violation; at $\delta_{CP} = 212^\circ$ even a flavor-universal coupling, $g\,U^TU$, has off-diagonal entries of 0.09–0.13 g. The Majorana phases drop out of every rate, and for a single coupled flavor f so does δ_{CP}, $|\lambda_{ij}| = g\,|U_{fi}||U_{fj}|$ (tests). A high-energy neutrino ν_i of energy E scattering on a relic ν_j at rest has $s = 2Em_j$. The cross sections of the appendix cover $\nu_i\nu_j \to \nu_k\nu_l$ through the s, t and u channels and their interference (Eqs. A11, A12), and, above $s = 4M^2$, double-scalar production $\nu_i\nu_j \to \phi\phi$ through neutrino exchange (A11, A13), each scalar decaying at once into two neutrinos with an energy spectrum flat in $[0, E_\phi]$ (A16); the width is $\Gamma = (M/16\pi)\sum_{kl} g^2_{kl}$ (A15). The secondaries' energies follow from $t = (p_i - p_k)^2$: $E_k = E(1 + t/s)$ for a neutrino and $E_A = E(1 + (t - M^2)/s)$ for a scalar.

Three readings of the appendix were settled by checks rather than assumed. The factor 1/2 of Eq. (A12) applies only to identical final states ($k = l$): with it, (A12) integrated over t and summed over the final states reproduces (A11) to 10^{-13}. The kinematic range of the scalar's t is $M^2 - s/2 \mp \sqrt{s(s/4 - M^2)}$, where $st + (t - M^2)^2$ vanishes; the appendix prints it without the factor $\sqrt{s}$ and the shift M^2. And the targets: the cross sections are summed over the relic's spin, and the appendix says to multiply them by half the number of targets, i.e. by the density per helicity state, $n = 3\zeta(3)T^3_\nu/4\pi^2 = 56$ cm^{-3} per mass state today. An independent narrow-width derivation agrees: an incident ν_L on an unpolarized relic has $\sigma = (\pi g^2/2)\,\delta(s - M^2)$ per relic, against 2×56 relics, while the appendix's s channel integrates to πg^2 across the resonance (a

test).

Complex couplings.—The appendix assumes real couplings; we generalized it from the helicity structure of the amplitudes. For an incident ν_i of helicity $-\frac{1}{2}$ and the relic at $-\frac{1}{2}$, the s channel ($\lambda_{ij}\lambda_{kl}$), t channel ($\lambda_{ik}\lambda_{jl}$) and u channel ($\lambda_{il}\lambda_{jk}$) lead to the same final state (both neutrinos at $+\frac{1}{2}$) and interfere; the s channel into two neutrinos at $-\frac{1}{2}$ ($\lambda_{ij}\lambda^*_{kl}$) and, with the relic at $+\frac{1}{2}$, the t channel ($\lambda_{ik}\lambda^*_{jl}$) and the u channel ($\lambda_{il}\lambda^*_{jk}$) add incoherently. Equation (A12) becomes the same expression with moduli in the squared terms and real parts in the interference: the s–t term carries $\mathrm{Re}[\lambda_{ij}\lambda_{kl}\lambda^*_{ik}\lambda^*_{jl}(s - M^2 - iM\Gamma)]$ and likewise s–u, the t–u term $\mathrm{Re}[\lambda_{ik}\lambda_{jl}\lambda^*_{il}\lambda^*_{jk}]$. The part with $M\Gamma$ changes sign between neutrinos and antineutrinos (whose couplings are conjugated): a CP asymmetry of relative size $\Gamma/M = \sum|\lambda|^2/16\pi$, which the transport, following both helicities together, averages out. A test integrates the generalized $d\sigma/dt$ for random complex couplings and recovers the closed-form channel sums below to 10^{-10}.

One formula we do not adopt. The amplitudes of double-scalar production, Eqs. (A3) and (A4), sum over the exchanged neutrino ν_k; with the neutrino masses neglected in its propagator the sum is coherent, and the rate carries $|\sum_k \lambda_{ik}\lambda^*_{kj}|^2 = |(\lambda\lambda^\dagger)_{ij}|^2$ (the exchanged neutrino connects λ at one vertex to λ^* at the other, the relic at the opposite helicity). The appendix prints $\sum_k g^2_{ik}g^2_{jk}$ in (A10), (A11) and (A13), the incoherent sum. The two agree for a flavor-universal coupling without CP violation; for a single flavor f the printed form is smaller by $\sum_k |U_{fk}|^4$, about 0.35 for ν_τ, so it fails for non-universal textures. We use the coherent form; the option `phiphi_coupling = "paper"` reproduces their figures (their Fig. 4 is consistent with the printed form).

Summed over the final states, the absorption of ν_a on the relic ν_r is

$$\sigma_{ar} = \frac{|\lambda_{ar}|^2 G_2}{16\pi}\frac{s}{B} + \frac{2H_{aa}H_{rr}N_2 + |H_{ar}|^2 N_5}{16\pi s} + \frac{2\,\mathrm{Re}(\lambda_{ar}Q_{ar})}{16\pi}\frac{(s-M^2)N_4}{B} + \frac{|H_{ar}|^2}{32\pi}I_{\phi\phi}, \quad (2)$$

with $H = \lambda\lambda^\dagger$, $Q = \lambda^*\lambda\lambda^*$, $G_2 = \sum_{kl}|\lambda_{kl}|^2$, $B = (s-M^2)^2 + M^2\Gamma^2$, $N_{2,4,5}$ the integrals over the secondary's energy fraction x of the profiles $(1-x)^2/(1-x+\mu)^2$, $(1-x)/(1-x+\mu)$ and $x(1-x)/[(x+\mu)(1-x+\mu)]$, $\mu = M^2/s$, and $I_{\phi\phi} = (s-2M^2)\,\Phi(w)/s^2$, $\Phi(w) = 2(3-w^2)\,\mathrm{artanh}\,w - 6w$, $w = \sqrt{s(s-4M^2)}/(s-2M^2)$. The secondaries follow the same couplings: for example the t channel gives ν_c with x-spectrum $|\lambda_{ac}|^2 H_{rr}(1-x)^2/[8\pi s(1-x+\mu)^2]$, and every $2\to2$ absorption yields two secondaries and every double-scalar one four (identities that a test checks for random real and complex couplings). The closed forms cancel catastrophically in two places, near the double-scalar threshold ($\Phi \sim w^5$) and far below the resonance ($N_2 \sim s^2/3M^4$); there we use the series and Gauss–Legendre of cancellation-free integrands, and every kinematic integral agrees with 50-digit arithmetic to 4×10^{-15} from threshold to $s/M^2 = 10^{14}$.

6.2 Dirac neutrinos and a vector mediator

On 2026-10-13 Mauricio asked that the physics of both be checked before any code; the checks (`tests/amplitudes_bruteforce.py`) evaluate every tree amplitude with explicit Dirac matrices and helicity spinors, Majorana fermions by the fermion-flow rules of Denner, Eck, Hahn & Küblbeck, independently of every closed form, and the choices below were decided after them.

The brute force reproduces what we took over.—Eq. (A12) for real couplings (10^{-15}), our complex-coupling form of it including the term odd under $\nu \leftrightarrow \bar\nu$ (a CP asymmetry of relative size Γ/M, which the transport averages), double-scalar production with the coherent factor (the printed one is off by factors 0.08–2), and the textbook Bhabha and Moller helicity amplitudes as a check of the engine itself.

Scalar, Dirac neutrinos.—With Dirac fields the coupling is $-\phi\, h_{ij}\bar\nu_i\nu_j$, $h = U^\dagger G U$ Hermitian: the right-handed neutrinos rotate like the left-handed ones (the paper's convention; nothing observable fixes their basis; decided). The scalar flips chirality, so every vertex joins a state that interacts weakly (ν_L, $\bar\nu_R$: visible) to a sterile one (ν_R, $\bar\nu_L$). The relics, at rest and

unpolarized, present both helicities in the collision, and the scalar couples to both: an incident ν_a meets $\bar{\nu}_r$ of helicity $-$ in the s and t channels (final states: one visible, one sterile), ν_r of helicity $-$ in the t and u channels (both sterile), and the opposite helicities in the t channel alone. Summed, the total is the appendix's Eq. (A8) exactly; the split into visible and sterile secondaries, per mass state and energy, was derived by hand and agrees with the brute force to 10^{-15}. The appendix's Table I, read with its stated factors, overstates the visible production by up to 3.5%; we do not use it. The sterile secondaries scatter too, and by parity an incident ν_R gives the secondaries of a ν_L with visibility swapped (checked), so the visible flux is $(S + D)/2$ from two runs of the transport, S with the kernel visible + sterile and D with visible $-$ sterile (decided: tracked; the option `"drop"` discards them, as 2107.13568 does). Width $M \sum |h|^2/8\pi$.

Vector mediator.—Coupled to left-handed currents (decided), $-V_\mu G_{\alpha\beta} \bar{\nu}_\alpha \gamma^\mu P_L \nu_\beta$, as in gauge models like $L_\mu - L_\tau$; mass basis $U^\dagger G U$ for Majorana and Dirac alike. Only the relic components that behave as ν_L (helicity $-$) and as $\bar{\nu}_R$ (+) interact; a Majorana relic offers both (56 + 56 per mass state), a Dirac one half of them, so Dirac rates are half the Majorana ones, and every secondary is visible. The brute force confirms, for complex couplings, $|\mathcal{M}|^2 = 4u^2|A_s/(s - M^2 + iM\Gamma) + A_t/(t - M^2)|^2$ for $\nu\bar{\nu} \to \nu\bar{\nu}$ ($A_s = G_{ra}G_{kl}$, $A_t = G_{ka}G_{rl}$), $4s^2|A_t/(t - M^2) + A_u/(u - M^2)|^2$ for $\nu\nu \to \nu\nu$, and $4|(GG^\dagger)_{ar}|^2[t/u + u/t + 4M^2 s/tu - M^4(1/t^2 + 1/u^2)]$ for $\nu\bar{\nu} \to VV$ (to 10^{-12}); the s channel gives the Z-burst $\langle\sigma\rangle = 2\sqrt{2}\pi G_F$ for the Standard-Model coupling. Each V decays at once into $\nu\bar{\nu}$; in the relic rest frame, where it is ultrarelativistic, a transverse one gives $3z^2 + 3(1 - z)^2$ for the pair and a longitudinal one $6z(1 - z)$ per neutrino. Its longitudinal state there is the light-cone polarization $k/M - Mp_2/(k \cdot p_2)$, and the Ward identity removes one diagram: $|\mathcal{M}_0|^2 = 8sM^2(t - M^2)^2/[t^2(u - M^2)^2]$ (derived by fitting the brute force, exact to 3×10^{-14}). The longitudinal fraction reaches 55% near threshold and $\sim 20\%$ for soft vectors at $s = 100M^2$ (decided: exact).

The forward peak.—Above the resonance the vector's t channel is forward-peaked: $\sigma \to |G|^4/4\pi M^2$, constant, where the scalar's falls as $1/s$ (Fig. 1). Nearly every such scattering leaves the neutrino in its own energy bin and gives $\sim M^2/2m_r$ to the recoiling relic, which becomes a neutrino near the resonance energy; with a flavor-dependent coupling it may also change the mass state. These are real, incoherent scatterings (the relic recoils). The transport removes from a bin's equations the returns that change nothing (complementary integrals, without cancellation), keeps every recoil, and solves the bin's three mass states exactly over the step with a matrix exponential, because the quasi-elastic conversions are fast enough to make them stiff (the scalar keeps its scheme, its returns being of order $\Delta \ln E$). Each return that changes nothing still costs the neutrino $\sim M^2/s$ of its energy, and at small M there are enough of them to move it by many bins (at 1 keV and $g = 0.03$, $\sim 10^7$ returns of 10^{-6} each; two thirds of the energy loss is in the returns, a third in the jumps out of the bin). The transport carries their losses as a drift in $\ln E$: an upwind flux to the bin below, in the daughter's state, at the rate of the returns times their mean shift $\int_{1/h}^1 (-\ln x) P(x)\, dx$ (closed form plus a bounded remainder, 10^{-7} against 40 digits), which the top-down sweep passes on within the step however fast it is. Without it (until 2026-10-13) the energy-averaged fractions of flavor-universal vector couplings at 1 keV and $g = 0.03$ moved by 7×10^{-3} between 401 and 1601 bins, as much as the effects measured; with it, 10^{-4} (3×10^{-5} at 801 bins), and ν_e couplings at 1 and 10 keV agree to 10^{-4} or better at 401 bins; where nearly all the flux is gone (10^{-4} of it left, ν_e at 100 keV and $g = 0.1$) to 10^{-3}, falling with the bin width. The scalar has no forward peak and no drift.

Far above the resonance this is all that happens: at $M = 1$ keV ($s/M^2 > 10^5$ in the band) a scattering moves the neutrino by $\sim M^2/s \sim 10^{-6}$ in energy but projects it onto the coupled flavor ($\sum_k G_{ka}|\nu_k\rangle \propto |\nu_e\rangle$ for ν_e couplings), at the rate $W_{a\to x} = 2\kappa\, n\, |G_{xa}|^2 \,\mathrm{Tr}(G^\dagger G)/4\pi M^2$ from both relic components. The flux at each energy then obeys $dF/dz = (1 + z)^2 K_0 F/H(z)$, $K_0 = W - \mathrm{diag}(\sum_x W_{x\cdot})$, solved by one matrix exponential per source redshift; the transport reproduces its flavor fractions to 10^{-6} (Majorana and Dirac, $g = 5 \times 10^{-4}$ and 10^{-3}; a test). W

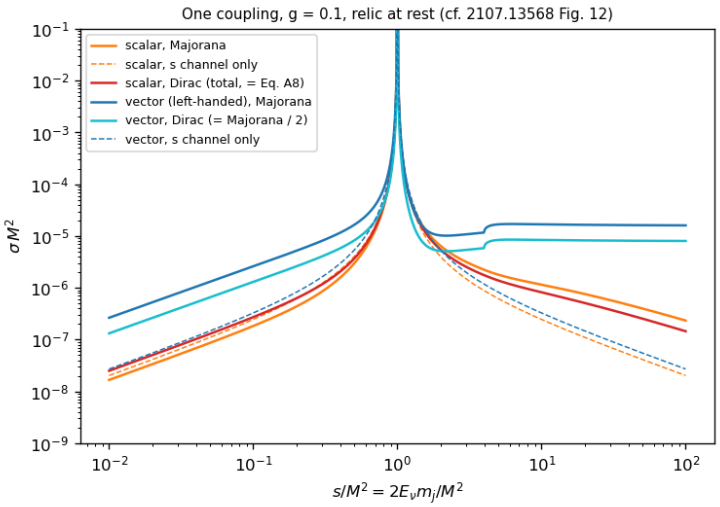


Figure 1: **Total cross sections** for one coupling, $g = 0.1$, relic at rest: scalar (Majorana, Dirac) and vector (Majorana; Dirac is half), with the s channels alone. The scalar Majorana curves reproduce 2107.13568, Fig. 12; all from the closed forms checked by brute force.

being symmetric, the conversions drive the flux toward equal mass states, 1/3 of each flavor at Earth: a shift of ~ 0.01 from the pion sources' $0.326 : 0.343 : 0.330$, which the 2050 projection resolves. The rate grows as g^4/M^2, so the limits below the resonant window follow $g \propto \sqrt{M}$ (Sec. 6.6).

6.3 Transport

Over cosmological distances the mass states decohere and there is no coherent potential to speak of (the relic neutrinos' own contribution to the neutrino masses through ϕ is negligible), so the transport follows the number of each mass state per unit observed energy, $F_a(E)$, on the solver's geometric grid; the physical energy at redshift z is $E(1+z)$, so redshift losses need no term of their own, and the sources inject $\Phi_0[E(1+z)]^{-p} R(z) \sum_\alpha |U_{\alpha a}|^2 f_{\alpha,S}$ per unit time, as for the dark portal. Both helicities of a mass state are followed together: with real couplings the interaction is CP-symmetric, so their sum evolves exactly whatever the source's $\bar\nu/\nu$ ratio (the solver returns it split in the injection's proportions; the split of the secondaries is not modelled).

Redshift steps are uniform in $\ln(1+z)$, one per energy bin by default (`z_substeps`), so that μ of bin j at the midpoint of step m, $\mu = (M^2/2E_0 m_r)\, e^{-(jq+m+1/2)\Delta}$, falls on one lattice per relic state: the exact bin integrals of the seven regeneration profiles are tabulated once per mediator and relic masses (not per coupling, so scans in g reuse them; ~ 30 MB at the analysis grid). Each step integrates the optical depth exactly: the Breit–Wigner terms in $\theta = \arctan[(s-M^2)/M\Gamma]$, in which the resonance is smooth however narrow (a crossing contributes $\int A(s)\, ds/B = M\Gamma^{-1} \int A\, d\theta$, with A the slowly varying density and Jacobian), the others by Gauss–Legendre in s. Then the bins are swept from high to low energy, each solving its equations over the step exactly with constant rates and sources, and passing down the number it absorbed: its own secondaries through a 3×3 system over the mass states, those of lower bins through the tabulated profiles. The number at $z = 0$ equals the injected number plus the secondaries kept in the grid minus the absorbed, to 10^{-14}. The grid extends tenfold above $E_{\max}$ so that what is injected above it cascades in. A run takes 0.3–1 s at the analysis grid (400 bins, 81 steps).

Table 5: **Checks of the secret-interaction module** (`tests/test_si.py`, `tools/si_validation.py` on Tycho). Deviations of fluxes are relative to the free-streaming flux at the same energy.

Check	Result
Channel sums vs Eq. (A11) as printed; vs quadrature of (A12) + (A13)	10^{-13} (A11 itself loses digits above $s/M^2 \sim 10^3$)
Kinematic integrals vs 50 digits	4×10^{-15}, threshold to $s/M^2 = 10^{14}$
Profile bin integrals vs 30 digits; their sums vs closed-form norms	1.5×10^{-14}; 10^{-14}, $\mu = 10^{-9}$ to 10^8
Double-scalar neutrino spectrum vs 30 digits	2×10^{-10}
No coupling vs free streaming	10^{-14}
Number balance	10^{-14}, every texture, $M = 1$ keV–1 GeV, g up to 1
Absorption alone vs quadrature of $e^{-\tau}$ along each path	median 10^{-12}–10^{-4}; up to 1.2% in the bins at a narrow resonance's sharp edge (one step per bin; 0.2% with four)
Bins and steps vs 1601 bins, four steps per bin (nested grids: no interpolation)	at 401 bins, spectra 2×10^{-3} pointwise (median 2×10^{-4}), up to 4% in the one bin below a very narrow resonance's edge ($g = 0.01$; first order in the bin width); energy-averaged fractions $\leq 2 \times 10^{-4}$ in all eight cases (scalar and vector, Majorana and Dirac)
2107.13568 Fig. 5	reproduced (Fig. 2)
2107.13568 Fig. 3 (Majorana, Dirac)	Majorana within 1–4%; Dirac 1% below the dips, 9% above the first dip (open; Fig. 3)
Tree amplitudes by brute force (explicit spinors; Majorana fermions by fermion flow)	(A12), its complex form, the Dirac visible/sterile split 10^{-15}; vector $\lvert\mathcal{M}\rvert^2$ 10^{-12}, longitudinal fraction 3×10^{-14}
Vector: tensors vs $\int d\sigma/dt$; narrow s channel vs the Z-burst normalization	10^{-10}; 2×10^{-4}
Dirac: absorption alone; tracked regeneration over Majorana's	equal to Majorana's (3 digits); 0.499–0.500
Vector far above the resonance vs its closed form (conversions at fixed energy)	flavor fractions 10^{-6}
Vector drift integrals vs 40 digits; fractions at 201 vs 401 bins (1 keV)	10^{-7}; 1.5×10^{-4} (5×10^{-3} without the drift)

6.4 Validation

The one error above the per-mille level is local and understood: within the step in which a bin crosses a narrow resonance, the particles injected during the step are treated as exposed to the step-averaged absorption, while they are in fact absorbed only if injected before the crossing. It is first order in the step and confined to the one or two bins at the dip's high-energy edge; energy-averaged fractions do not see it (10^{-5}). Splitting the crossing step at the crossing would remove it, if the event-rate analyses need it.

Figure 2 repeats Fig. 5 of 2107.13568 (ν_τ couplings, $M = 5$ MeV, $\Sigma m_\nu = 0.1$ eV in normal ordering, an E^{-2} flux): the two dips at $E_{\rm res} = M^2/2m_j$, the bump of the regenerated neutrinos below them, the slow recovery above the resonance at $g = 0.1$ (where the t and u channels and double-scalar production dominate), and the narrow-width approximation that does not depend on g once $g \gtrsim 10^{-3}$. Our full computation at $g = 0.1$ peaks $\sim 5\%$ above theirs below the dips; with their (printed) double-scalar coupling the difference halves, the rest being within the differences of source evolution and cosmology.

Figure 3 repeats the bottom panel of their Fig. 3, the same model at $g = 0.01$ with Majorana

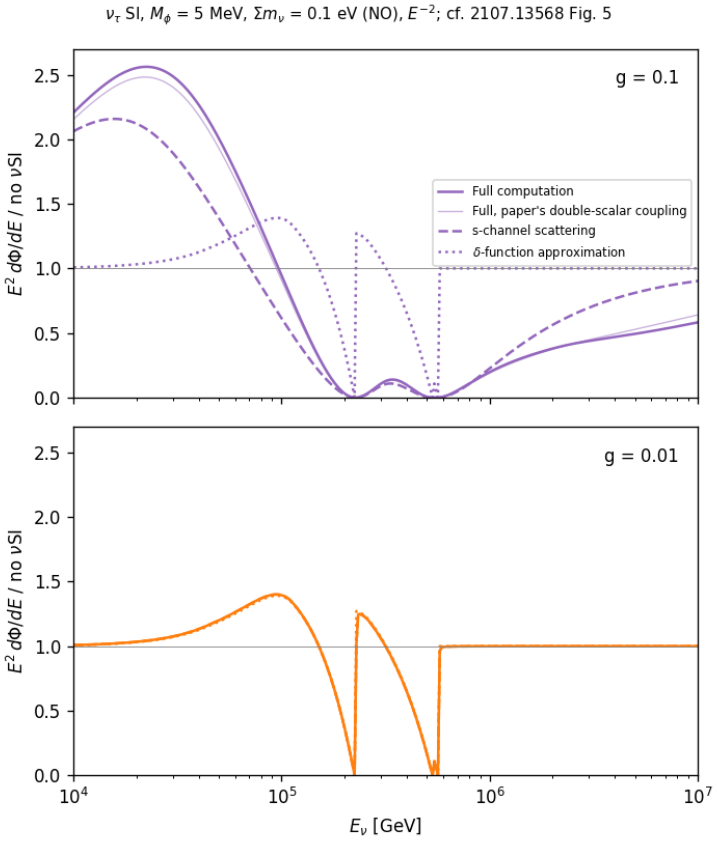


Figure 2: **Fig. 5 of arXiv:2107.13568 recomputed** with `src/dp_si.py` (all-flavor flux over the flux without interactions; ν_τ couplings, $M = 5$ MeV, $\Sigma m_\nu = 0.1$ eV in normal ordering, E^{-2} sources up to $z = 4$; 400 bins). Thin line: the double-scalar coupling as printed in the paper. `tools/si_validation.py fig5`.

and with Dirac neutrinos. Read off their figure, the Majorana curve peaks at 1.42 below the dips (ours 1.40) and at 1.21 just above the first dip (ours 1.25); the Dirac curve, with the sterile secondaries dropped as they do, peaks at 1.11 below the dips (ours 1.12) but only 0.91 above the first dip, where ours gives 1.00. Our curves obey two exact relations: absorption alone is the same for both kinds of neutrino (to three digits at every energy), and with the sterile secondaries tracked the Dirac regeneration is exactly half the Majorana one (0.499–0.500 of it at every energy), as the parity mirror of Sec. 6.2 requires; dropping them keeps 0.42–0.47 of it. Their Dirac curve would need ~ 0.3 just above the first dip against ~ 0.47 below the dips, which no single treatment gives; the source of this one difference is open (their code, nuSIprop, would settle it). The source composition plays no role here: with $1:1:1$ (equal mass states, as theirs) or $1:2:0$ the curves agree to three digits, every state being absorbed in the dips.

6.5 The flavor-triangle planes

We ran the flavor-triangle analysis of Sec. 5 with the secret interactions in place of the dark portal: the projected 2050 measurement for pion sources, the projection's mixing, $m_{\text{lightest}} = 0.01$ eV, $E^{-2.5}$ sources up to $z = 4$, the whole sky (no Earth effect here), and planes in (M, g) for four textures, ν_e, ν_μ or ν_τ couplings alone and flavor-universal ones, and three non-universal textures of two flavors, g on ν_e and ν_μ, ν_e and ν_τ, or ν_μ and ν_τ (49×49 points from 1 keV to 1 GeV and $g = 10^{-6}$ to 1, read per mass at $\Delta\chi^2 = 3.84$; Tycho, commits `fe8e2dc` for the single flavors and `a6570a6`, with the couplings $U^T G U$, for the others; 4–9 minutes each on 16 cores;

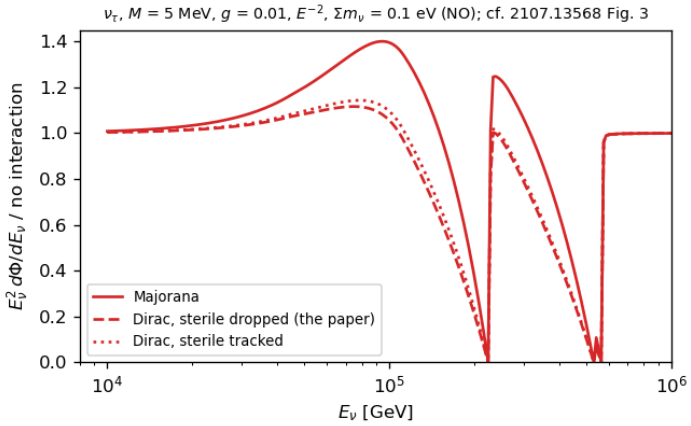


Figure 3: **Fig. 3 (bottom) of arXiv:2107.13568 recomputed**: Majorana neutrinos, and Dirac neutrinos with the sterile secondaries dropped (as the paper does) or tracked; ν_τ couplings, $M = 5$ MeV, $g = 0.01$, otherwise as Fig. 2. `tools/si_validation.py fig3`, `tools/plot_si_summary.py fig3`.

configurations `pion_2050_case_5_si_*.json`). Every mass column reaches the region without an effect: $\Delta\chi^2 = 0.00$ at $g = 10^{-6}$ (a first run starting at $g = 10^{-4}$ did not, near 1 MeV). Figure 4 shows the planes and Fig. 5 how the composition moves.

The limits have three regimes. Below ~ 200 keV every resonance lies below the band and the non-resonant channels act alone, with $\sigma \sim g^4/8\pi s$ independent of M: the limit is flat, $g < 6 \times 10^{-3}$–8×10^{-3} (ν_e), 8×10^{-3}–10^{-2} (ν_μ, ν_τ), 5×10^{-3}–8×10^{-3} (universal), 6×10^{-3}–10^{-2} for the pairs with ν_e. Between ~ 0.5 and 3 MeV the resonances $E_{\text{res}} = M^2/2m_j$ fall at 10–50 TeV, at the bottom of the band, where the $E^{-2.5}$ flux dominates the energy-averaged fractions, and the limits drop by two to three decades: $g < 1.7 \times 10^{-5}$ (ν_e, at 0.56 MeV), 3.1×10^{-5} (ν_μ, 0.56 MeV), 9.0×10^{-5} (ν_τ, 0.75 MeV), 8.8×10^{-6} (universal, 0.56 MeV), and 1.5×10^{-5}, 1.1×10^{-5}, 1.9×10^{-5} for the pairs $e\mu$, $e\tau$, $\mu\tau$. Above, the resonances move above most of the flux and the limits weaken: at 10 MeV $g < 0.15$, 0.32, 0.36, 0.075 and 0.10, 0.08, 0.14; above 30 MeV only the universal coupling and the $e\tau$ pair are still limited ($g < 0.8$ and 0.9 at 100 MeV).

Two features follow from flavor fractions being the only observable. Flavor-blind absorption does not change the composition: universal couplings act on it through the mass-dependent resonance energies and through the off-diagonal couplings of $g\,U^T U$. With $g \not\parallel$, as the paper would have it, the effects of the m_3 and the $m_{1,2}$ resonances compensated near 1 MeV and left an allowed gap ($g < 9 \times 10^{-3}$ there); the off-diagonal couplings close most of it (2.4×10^{-4} at 1 MeV), and elsewhere the two agree to 1% (median). And as g grows the composition can pass back through the projection's centre: thin allowed islands and channels run through the excluded regions (the widest, for the $\mu\tau$ pair near 100 keV, raises the limit read there to 0.13; Fig. 5 shows the paths, nearly straight lines through the standard point). Spectral data will remove both; they will also exclude much of the plane outright, since at $g \gtrsim 10^{-2}$ below 1 MeV more than 99% of the flux below 10^7 GeV is absorbed, which normalized fractions do not see. The resonant window overlaps the region below ~ 2 MeV that Big-Bang nucleosynthesis already disfavors (2107.13568, Fig. 1).

6.6 Four models

We repeated the four planes of Fig. 4 for the scalar with Dirac neutrinos and for the vector with Majorana and with Dirac neutrinos (configurations `pion_2050_case_5_si_{dirac,vector,`

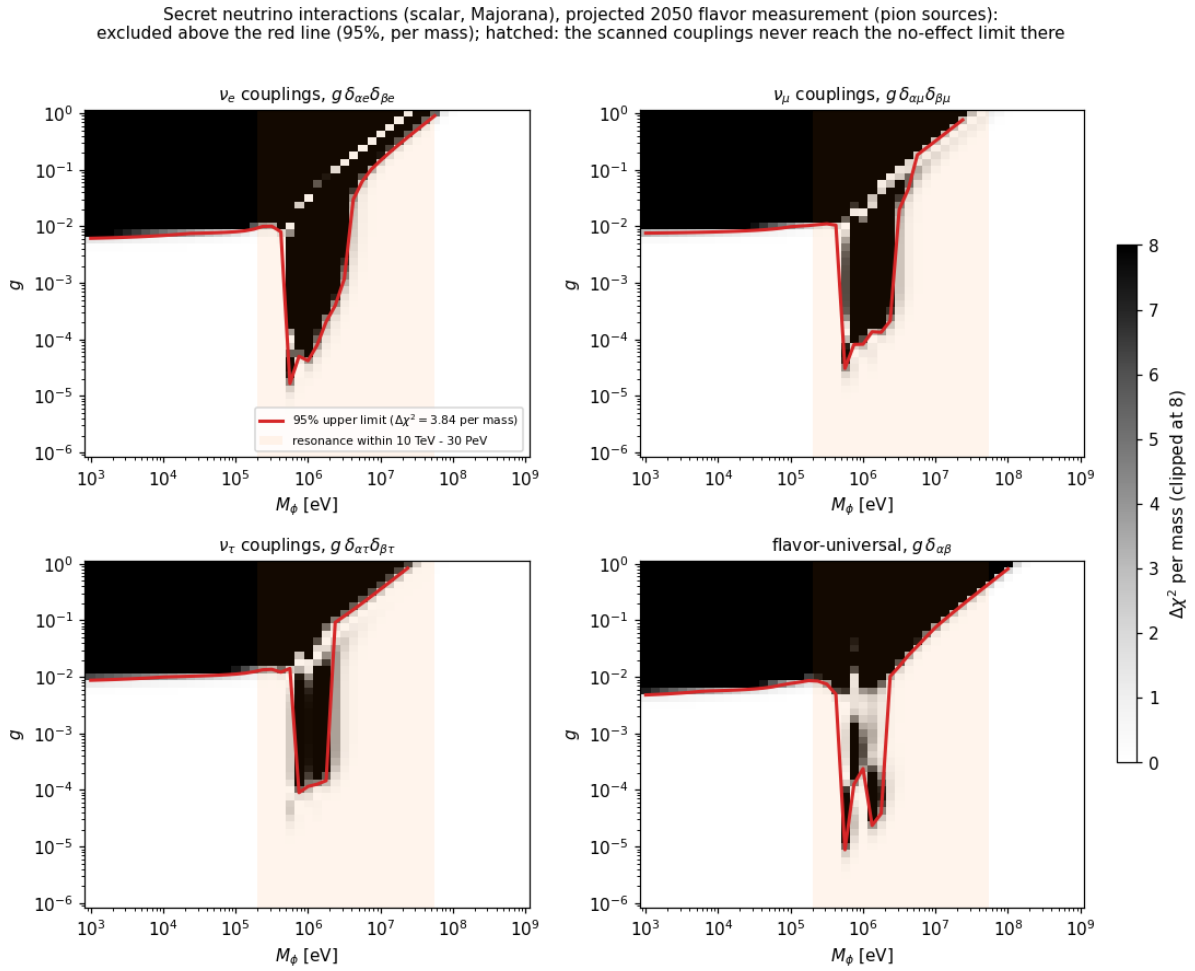


Figure 4: **Secret-interaction exclusion planes** of the projected 2050 flavor measurement (pion sources; scalar mediator, Majorana neutrinos): $\Delta\chi^2$ per mediator mass and the 95% upper limit (red; the top of each column's lowest allowed interval). Shaded: masses whose resonance falls in 10 TeV–30 PeV for some $z \leq 4$. `tools/plot_si_summary.py planes`.

`vector_dirac}_*.json`, the same grid; Tycho jobs 66302667 (commit `99a8ccc`) and 66304700–03 (`d82231c`, with the drift); 8–17 minutes per plane on 16 cores, the Dirac scalar the slowest, with two transport runs per point; the vector Dirac planes are in `output/figures/si_planes_vector_dirac.png`). Figure 7 shows one benchmark under the four models, Fig. 8 and Table 6 the limits, and Figs. 9 and 10 two of the models' planes.

The resonant window (0.5–3 MeV) gives the strongest limits in all four, within a factor ~ 1.5 of each other at its best mass (absorption alone being the same for Majorana and Dirac neutrinos, Sec. 6.4). For the Dirac scalar the excluded region inside the window breaks into islands (Fig. 10), where the composition's path passes back through the projection's centre, as it does more rarely for Majorana neutrinos (Sec. 6.5); the per-mass reading then jumps to the top of the lowest allowed interval (ν_τ: 2.2×10^{-4}, against 9.0×10^{-5} for Majorana).

Below the window the vector's quasi-elastic conversions (Sec. 6.2) set the limit for the single-flavor textures, $g \propto \sqrt{M}$: 4.9×10^{-4} for ν_e at 1 keV, twelve times below the scalar's 6.2×10^{-3}, with the flux intact. Their effect saturates at equal mass states, and where that shift is no longer enough (20–50 keV for ν_e) the limit comes from hard scatterings, as for the scalar. Flavor-universal couplings stay diagonal in the mass basis for the vector and for the Dirac scalar, $U^\dagger(g\not{K})U = g\not{K}$: no conversions, and the 1-MeV gap of $g\not{K}$ that $U^T U$ closes for the Majorana scalar (Sec. 6.5) remains for all three ($g < 0.009$–0.011 there, against 2.4×10^{-4}).

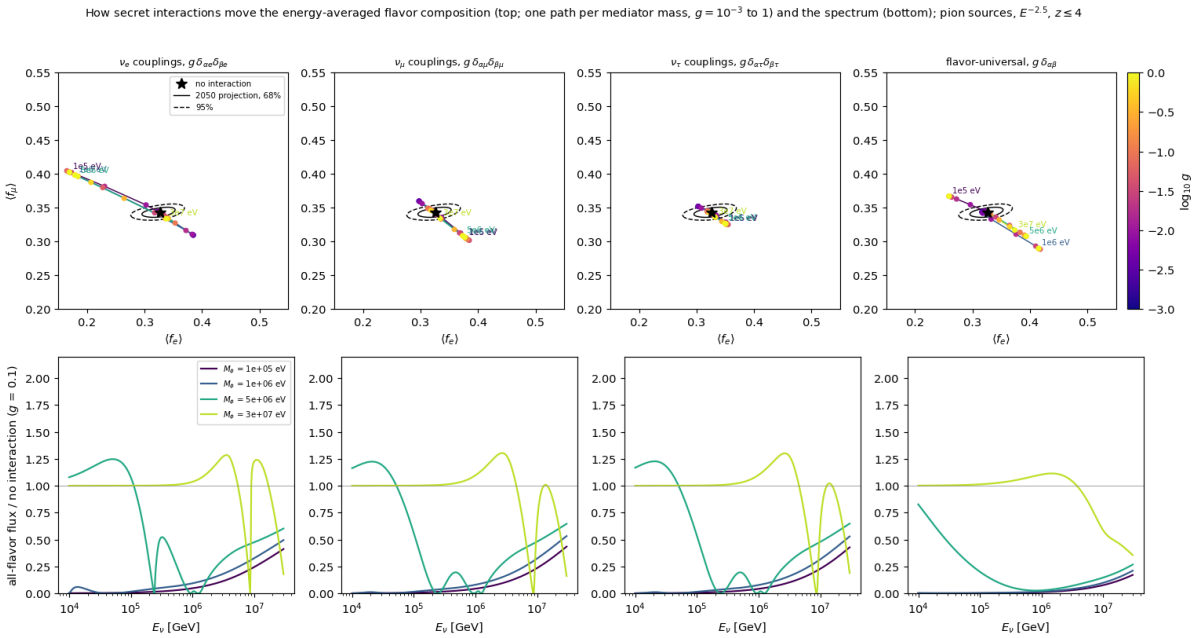


Figure 5: **How the composition moves.** Top: energy-averaged (f_e, f_μ) at the detector as g goes from 10^{-3} to 1 at four mediator masses, against the projection's 68% and 95% contours. Bottom: the all-flavor flux at $g = 0.1$ over the flux without interactions. `tools/si_flavor_paths.py` (Tycho), `tools/plot_si_summary.py paths`.

Below 400 keV universal vector couplings are not limited at all: the energy loss of their forward-peaked scatterings can remove the whole flux from the band (to 3×10^{-4} of its number at 1 keV and $g = 0.03$) while hardly moving its composition (f_e by -0.016, $\Delta\chi^2 < 3.84$), which normalized fractions cannot see. Spectral data will exclude that region outright. (Before the drift of Sec. 6.2, it appeared excluded above $g \simeq 4 \times 10^{-3}$: an artifact of the missing energy loss.)

Above the window the vector is limited further, because its cross section stays constant above the resonance where the scalar's falls as $1/s$: $g < 0.08$ for ν_e at 10 MeV (scalar 0.15), and for ν_e a limit up to 100 MeV (0.81). Dirac rates of the vector are half the Majorana ones, and its limits are weaker by $2^{1/4} \simeq 1.2$ wherever one process dominates (1 keV: 5.8×10^{-4} against 4.9×10^{-4} for ν_e).

Not yet included.—The relic neutrinos' thermal momenta (relics are at rest here, as in both papers; this matters for a nearly massless lightest state, whose relics are relativistic), the timing within the crossing step (Sec. 6.4), and spectral data, which would exclude outright the regions where the flux is absorbed without changing its composition.

7 Planned work

7.1 Event rates at IceCube

We will compute the event distributions in reconstructed energy for two event classes, from public IceCube responses, in a new package (`IceCubeRates/`).

Northern-Hemisphere muons.—Through-going muon tracks from below the horizon, where the Earth removes the atmospheric muon background. The response is IceCube's IceTracks-DR2 release (2008–2022, doi:10.7910/DVN/MMIIZA), which provides per-season muon-neutrino effective areas and smearing tables. For each bin of true energy and declination, a smearing table gives the fraction of events in each bin of reconstructed muon energy, angular distance

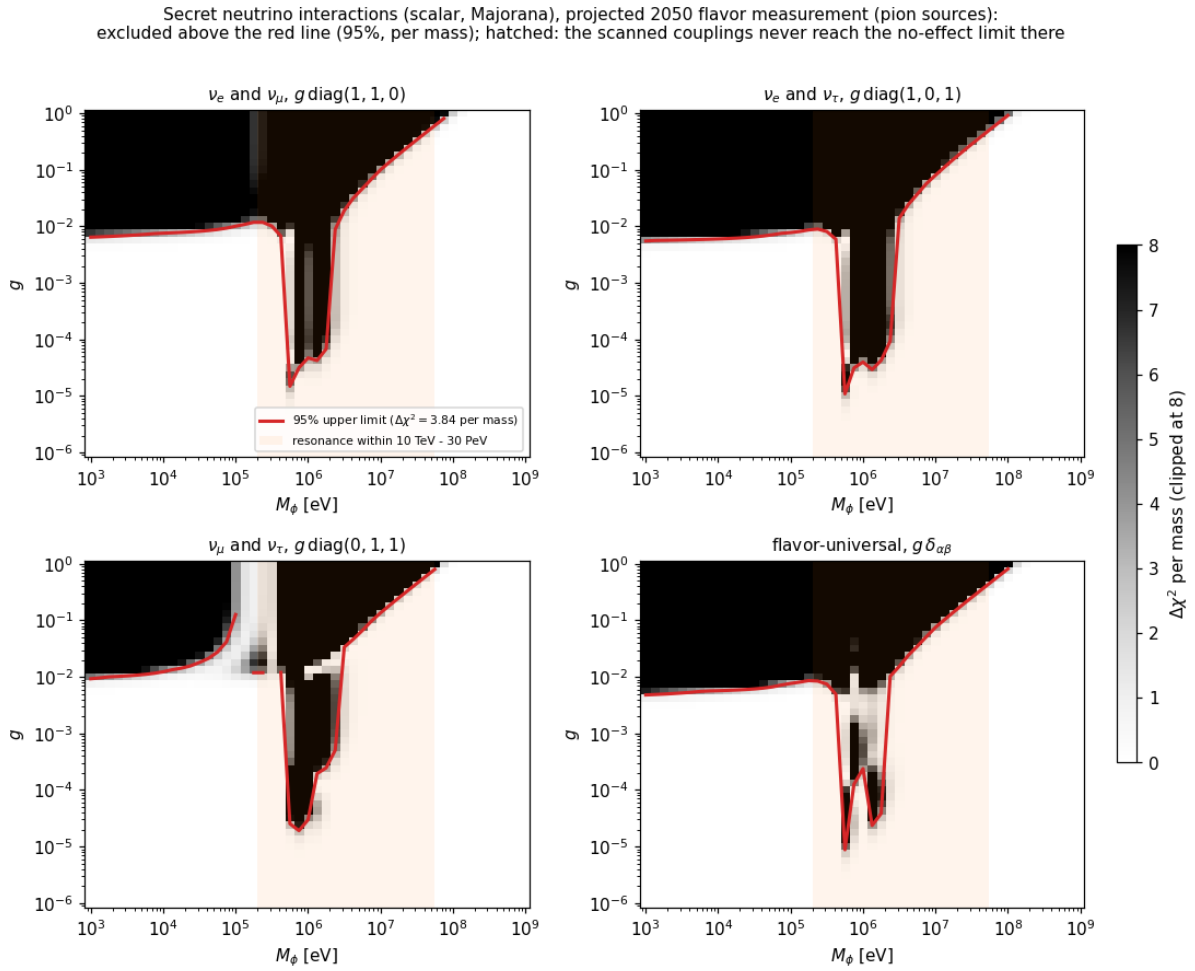


Figure 6: **Non-universal textures of two flavors**, as Fig. 4: g on ν_e and ν_μ, ν_e and ν_τ, ν_μ and ν_τ (complex mass-basis couplings $U^T G U$ at $\delta_{CP} = 212°$), and the universal coupling again. `tools/plot_si_summary.py planes2`.

from the true direction (point-spread function) and estimated angular error. Its 41 declination bins replace the 3 of the 2018 release, which matters near the horizon. We fold the flux with the PLEnuM framework (github.com/PLEnuM-group/Plenum), which also places detectors at other sites; our adapter converts the DR2 tables into PLEnuM's grids.

Starting events.—Cascades, starting tracks and double cascades, from IceCube's HESE 7.5-year release (github.com/icecube/HESE-7-year-data-release). It is the only public simulation of individual starting events that resolves all three morphologies. Each simulated event carries its flavor (neutrino or antineutrino), its interaction (charged current, neutral current or Glashow resonance), its reconstructed deposited energy and its morphology. It also carries a weight per unit flux, together with the atmospheric fluxes and self-veto probabilities at its energy and direction. We reweight each simulated event by our flux. For IceCube-Gen2 we scale these weights by the ratio of the Gen2 and IceCube effective areas published for the Gen2 design (doi:10.7910/DVN/XANWAT), per morphology, flavor, energy and zenith angle. Those tables carry no energy resolution, so the IceCube response supplies it.

Earth absorption and direction.—Both responses already contain absorption in the Earth, computed by IceCube with its nominal cross sections. We checked this in the DR2 tables: at 1 PeV the effective area for straight up-going neutrinos is 20 times smaller than at the horizon, and at 10 PeV 120 times smaller, while at 10 TeV it is nearly uniform. Our solver computes the other effect of the Earth, the flavor change caused by the dark-portal potential, which involves

Table 6: **95% upper limits on g of the four models** (projected 2050 flavor measurement, pion sources; per mass at $\Delta\chi^2 = 3.84$, the top of each column's lowest allowed interval). S/V: scalar/vector; M/D: Majorana/Dirac; best: the strongest limit and its mass in MeV; –: no limit up to $g = 1$ (or the column never reaches the no-effect limit). `output/analysis/`.

Texture	Model	Best (MeV)	1 keV	100 keV	10 MeV	100 MeV
ν_e	S, M	1.7×10^{-5} (0.56)	6.2×10^{-3}	8.0×10^{-3}	0.15	–
	S, D	1.9×10^{-5} (0.56)	5.9×10^{-3}	7.6×10^{-3}	0.23	–
	V, M	1.4×10^{-5} (0.56)	4.9×10^{-4}	6.0×10^{-3}	0.08	0.81
	V, D	1.9×10^{-5} (0.56)	5.8×10^{-4}	7.5×10^{-3}	0.096	0.98
ν_μ	S, M	3.1×10^{-5} (0.56)	7.6×10^{-3}	9.9×10^{-3}	0.32	–
	S, D	3.9×10^{-5} (0.56)	7.6×10^{-3}	9.2×10^{-3}	0.17	–
	V, M	2.3×10^{-5} (0.56)	9.1×10^{-4}	5.1×10^{-3}	0.15	–
	V, D	3.3×10^{-5} (0.56)	1.1×10^{-3}	6.1×10^{-3}	0.19	–
ν_τ	S, M	9.0×10^{-5} (0.75)	8.7×10^{-3}	0.011	0.36	–
	S, D	2.2×10^{-4} (0.75)	8.6×10^{-3}	0.011	0.2	–
	V, M	6.2×10^{-5} (0.75)	8.1×10^{-4}	5.9×10^{-3}	0.18	–
	V, D	8.8×10^{-5} (0.75)	9.6×10^{-4}	7.0×10^{-3}	0.21	–
universal	S, M	8.8×10^{-6} (0.56)	4.8×10^{-3}	7.7×10^{-3}	0.075	0.81
	S, D	1.2×10^{-5} (0.56)	5.7×10^{-3}	7.8×10^{-3}	0.076	0.81
	V, M	6.1×10^{-6} (0.56)	–	–	–	–
	V, D	8.6×10^{-6} (0.56)	–	–	–	–

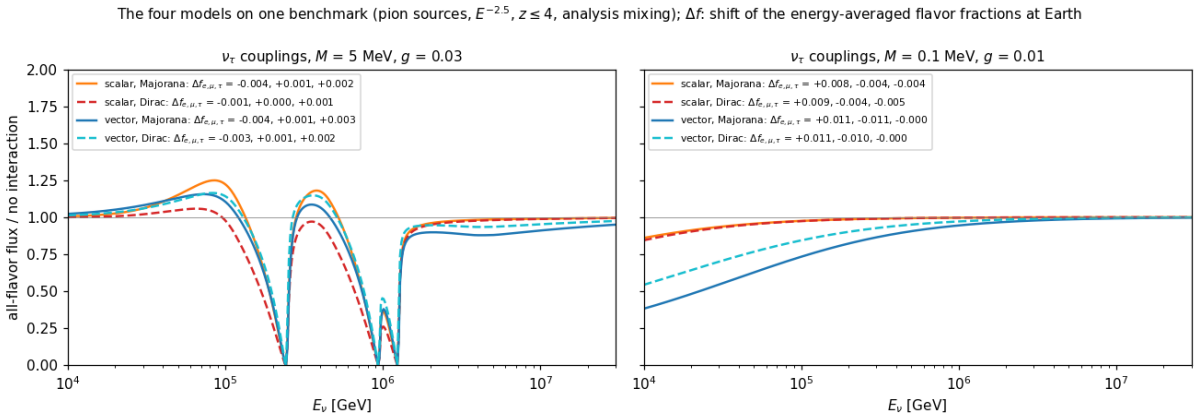


Figure 7: **One benchmark under the four models**: all-flavor flux over the flux without interactions, ν_τ couplings, and the shift of the energy-averaged flavor fractions at Earth. Left: $M = 5$ MeV, $g = 0.03$ (the resonant dips; above them the vector's constant cross section keeps acting). Right: $M = 0.1$ MeV, $g = 0.01$ (resonances below the band: the scalar's non-resonant channels against the vector's forward peak). `tools/si_validation.py models`, `tools/plot_si_summary.py models`.

no cross section. Because collisions remove all flavors nearly equally, the two act as independent factors along each path; the Glashow resonance, ν_τ regeneration and neutral-current scatterings that continue at lower energy break this slightly. Since both factors depend on direction, the solver must return the flux per Earth chord, and must add down-going directions, instead of the average it returns now.

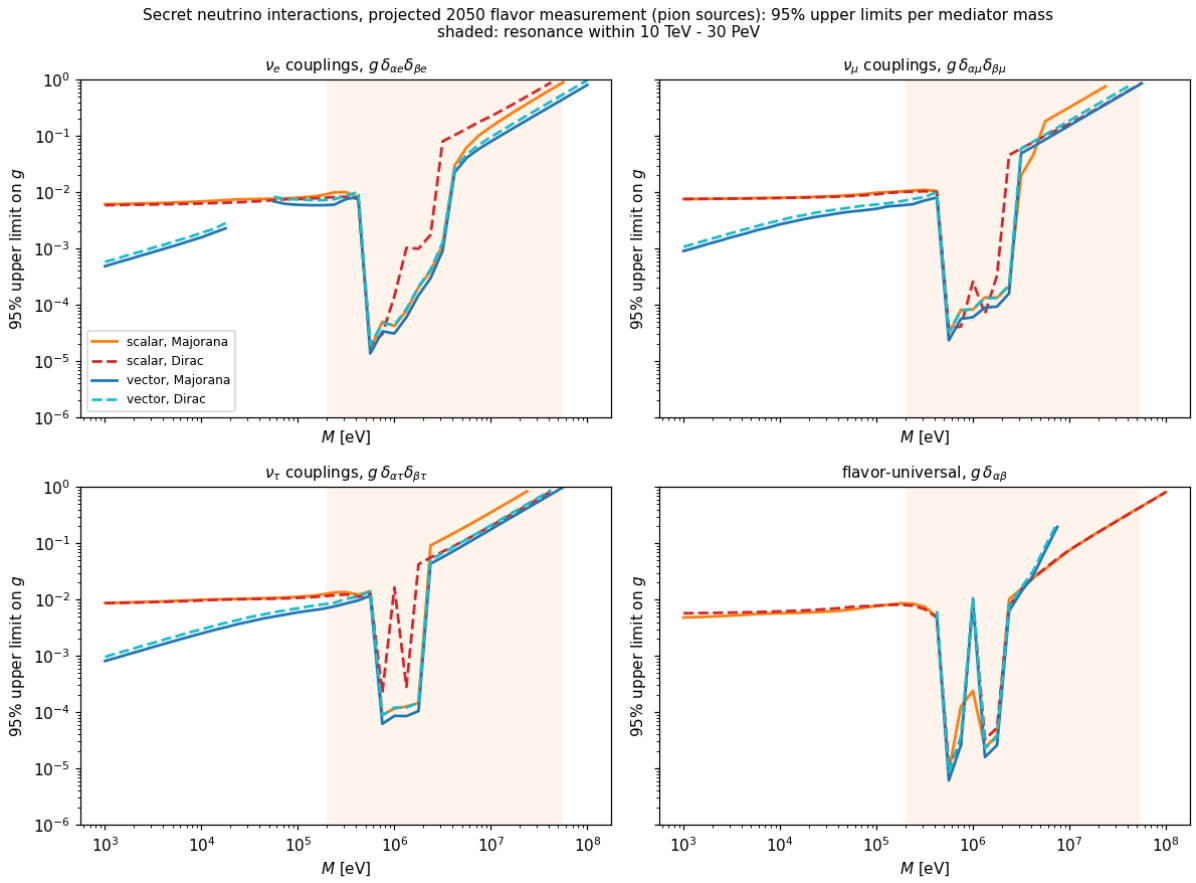


Figure 8: **95% upper limits of the four models** per mediator mass and texture (as Fig. 4). Gaps: masses with no limit up to $g = 1$, or whose lowest allowed interval is open. `tools/plot_si_summary.py limits`.

7.2 Systematic uncertainties

The fits will vary the following, agreed on 2026-10-12. *Astrophysical flux:* its normalization and spectral index. *Source composition:* $f_{e,S}$, with $f_{\tau,S} = 0$. *Atmospheric fluxes:* the conventional normalization, a tilt of the cosmic-ray spectrum and the twelve hadron-production parameters of Barr et al. above 30 GeV (pion regions G, H, I and kaon regions W, Y, Z, each for both charges). We compute their effects with MCEq (SIBYLL-2.3d and the H3a cosmic-ray model), exactly as in the NuFlavorSingleSource project. The prompt and atmospheric-muon normalizations and the self-veto vary as well. *Mixing parameters:* all six, constrained by the joint $\Delta\chi^2$ of NuFIT 6.1 with atmospheric data, per ordering. We use the three-dimensional table over $(\sin^2\theta_{23}, \Delta m^2_{31}, \delta_{\mathrm{CP}})$, the two-dimensional table over $(\sin^2\theta_{12}, \Delta m^2_{21})$ and the one-dimensional table over $\sin^2\theta_{13}$, so that each parameter appears in exactly one table. *Lightest neutrino mass:* free, with a prior on the sum of the masses.

The prior on the mass sum.—As in the GeoneutrinoDecay project, our prior is a half-Gaussian on the sum, which we derive from the lightest mass and the splittings. It adds $(\Sigma m_\nu/\sigma)^2$ to the χ^2, with $\sigma = L/\Phi^{-1}(0.95)$ for a published 95% upper limit L. We use $L = 0.055$ eV, from the cosmic microwave background (Planck, ACT and SPT-3G) with the baryon acoustic oscillations of DESI DR2 in ΛCDM, with a prior on Σm_ν that starts at zero (SPT-3G D1, arXiv:2506.20707, Eq. 96). Then $\sigma = 0.0334$ eV. The smallest sum allowed in normal ordering, 0.0587 eV, costs $\Delta\chi^2 = 3.08$; in inverted ordering, 0.0995 eV costs 8.85. Both orderings are fitted. As a check, this prior excludes the two minimal sums at 96.0% and 99.9%, against 96.6% and 99.9% stated in the paper.

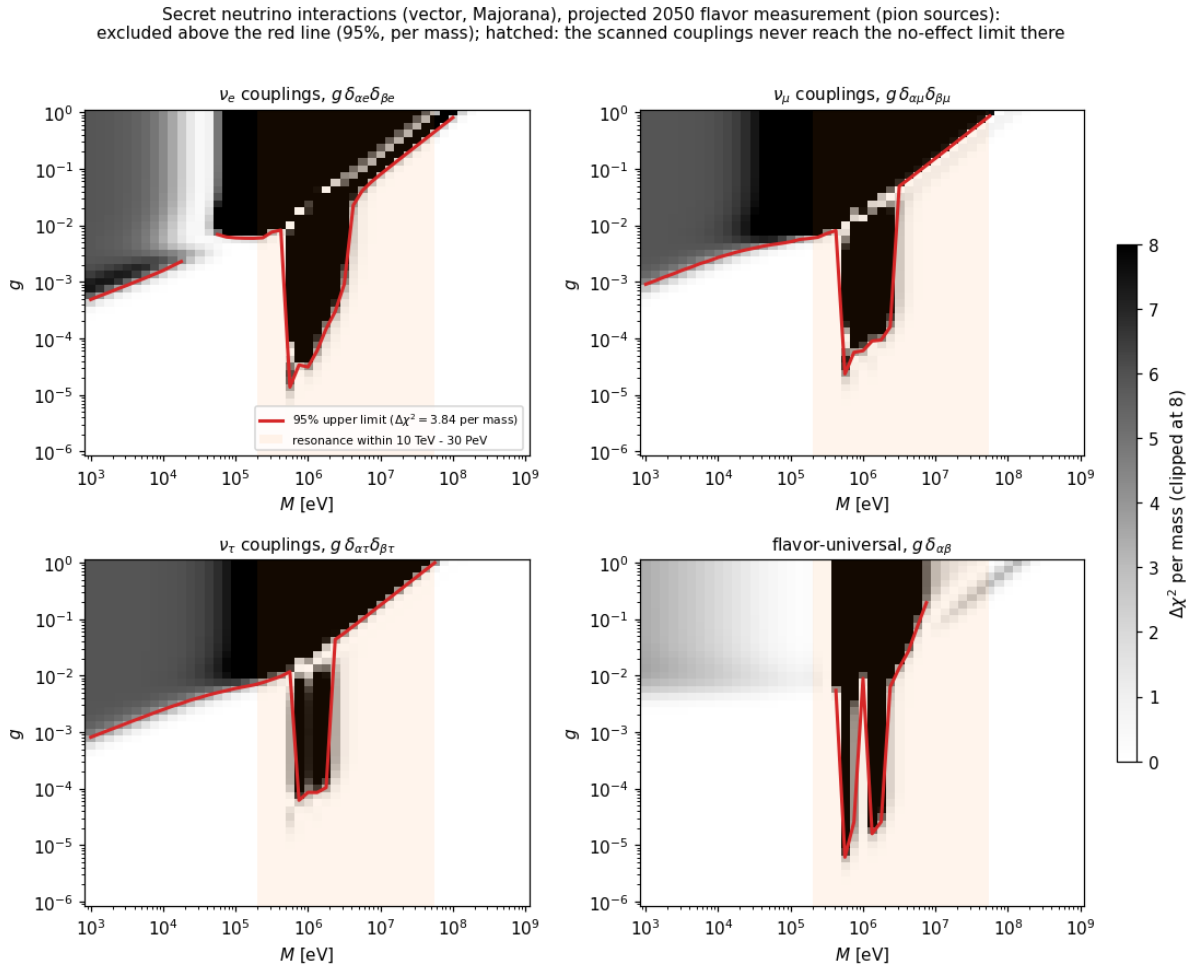


Figure 9: **Exclusion planes of the vector mediator** (Majorana neutrinos), as Fig. 4. Below the window, the quasi-elastic conversions ($g \propto \sqrt{M}$) for single flavors. `tools/plot_si_summary.py planes`, tag `vector_`.

Not varied.—The cross sections are fixed by the public responses; varying them would require generating our own effective areas. The detector-systematic corrections of the HESE release (optical efficiency, hole ice, ice anisotropy) are deferred, because they need the PHOTOSPLINE package.

7.3 Statistical soundness

We will establish that the analysis is sound in the order used in the GeoneutrinoDecay project. *Closure.*—Each response must reproduce a published IceCube fit before it is used for new physics; the HESE release includes its own power-law fit. *Injection and recovery.*—On Asimov data with known dark-portal parameters, and without a signal, the fit must recover the truth without bias. *Fits.*—Then the fits with the current profile-likelihood procedure. *Coverage.*—We generate pseudo-experiments with Poisson-fluctuated counts and fit each exactly like the data, 200 per point (which gives ±2% on a 10% rate). The points are the no-signal hypothesis, where we compare the test statistic with its asymptotic distribution, and the expected limit and half a decade beyond it. We add the parameter boundaries, the region where the mediator mass and the lightest mass trade off in setting the resonance energy, and the low-count corners (double cascades, highest-energy tracks). The pseudo-experiments change only the data, so the solver predictions on the scan grid are computed once and reused. *Decision.*—If no point under-covers,

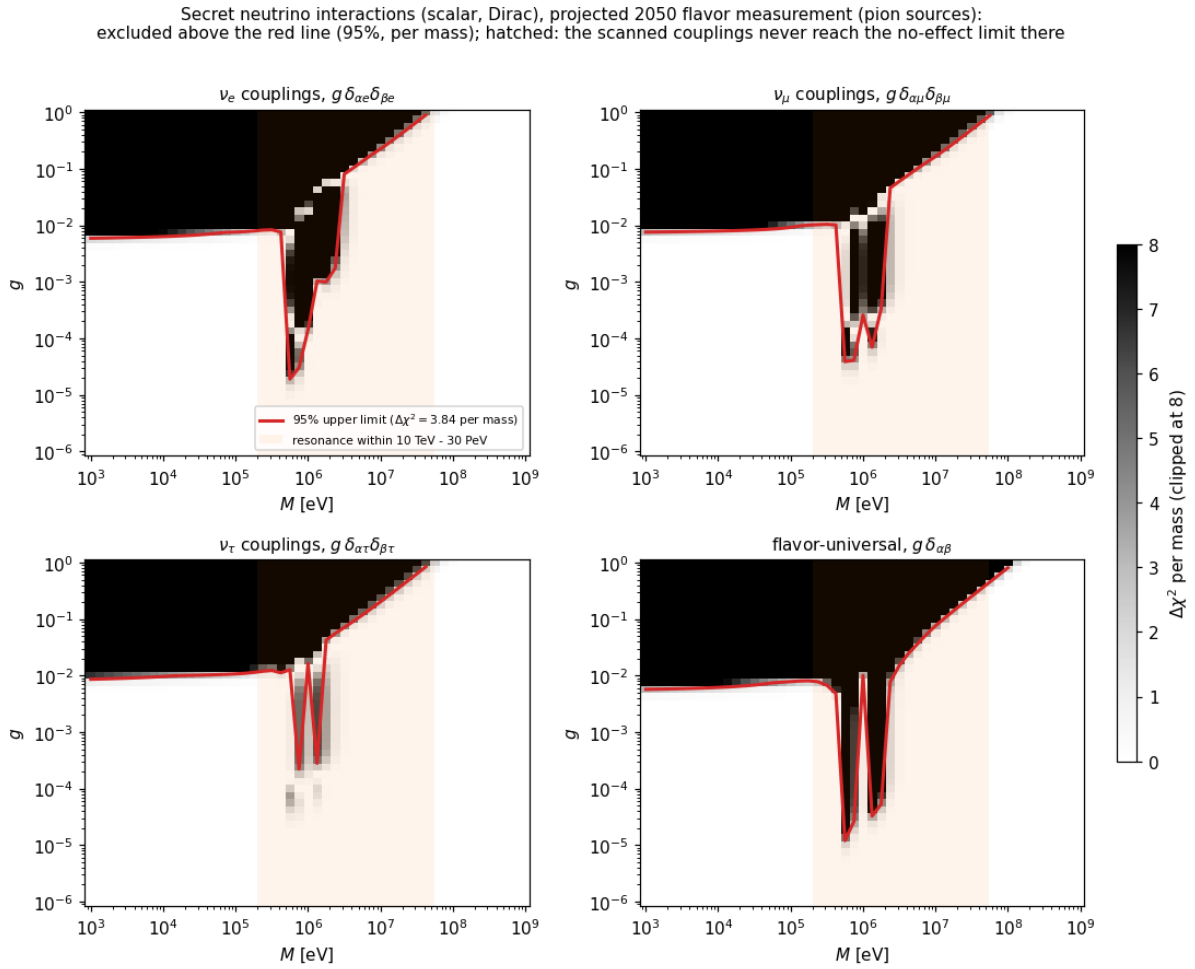


Figure 10: **Exclusion planes of the scalar with Dirac neutrinos** (sterile secondaries tracked), as Fig. 4: inside the resonant window the excluded region breaks into islands. `tools/plot_si_summary.py planes`, tag `dirac_`.

we keep the profile likelihood; otherwise we construct Feldman–Cousins intervals with PyFC (github.com/mbustama/FeldmanCousins), whose likelihood also accounts for the finite size of the HESE simulation.

The projected flavor analysis.—It has no counts to fluctuate; its pseudo-experiments must shift the centre of the projected measurement by its own width, which is an approximation we will state with its results.

7.4 Computation

Our heavy runs go to Tycho, the cluster of the Niels Bohr Institute, set up as in the GRBsim and NuFlavorSingleSource projects. The repository and a conda environment live on the shared file system, and the job scripts read their core count from SLURM. Jobs take 16 cores by default and up to 32. Before the first run, the solver must size its process pools from the cores the job was given; it now uses the core count of the whole node, so a 16-core job would start 127 processes. The first job will repeat the benchmark validation of Sec. 4 against the laptop's results.

8 Open items

Residual in the phase-noise regime (resolved).—At "cosmoX2" the chord count moved the averaged fractions by $\sim 6 \times 10^{-4}$, and the Earth step count by 2.9×10^{-3}. We attributed this to phases that double precision cannot follow; it was the propagator's non-unitarity. Since 2026-10-12 the results with 12, 24 and 50 chords, and with 8, 25 and 100 Earth steps, agree to 10^{-14} (Appendix A.10).

Analyses to rerun.—Profiles and limits computed before 2026-10-12 used a solver whose cascade ran away and whose absorption did not depend on the couplings (Appendices A.8 and A.9); they are void, not merely shifted. The new analysis package replaces the old pipeline, and its first production run is planned for Tycho.

Effect sizes to re-measure.—The effects quoted in Appendices A.1 to A.7 were measured with the solver of the time, before the corrections of Appendices A.8–A.10. They describe the change each correction made then; their size on the corrected solver is not known. **[Re-measure]**

Run time.—The production defaults ran in 0.32 s on 8 cores on 2026-07-29; on 2026-10-12 the same machine took 0.80 s before the solver corrections (commit `135ed02`) and 0.71 s after them (1.04 s on one core). **[VERIFY: find what changed between July and `135ed02`]**

Thermal-rate tails.—Far from resonance, where the rates are 10^{-6} to 10^{-10} of their peak, the semi-analytic kernel's results are dominated by rounding: a change in how numba compiles it (2026-10-13, numpy's error model for the fastmath kernels) moved them by up to 70%, while within 10^{-3} of the peak they agree to 3×10^{-12}. At "strong" this moves the flux in its absorption dip by 7×10^{-6} and the flavor fractions by 5×10^{-10}; the same happens between machines. **[Check whether those cells' optical depths matter anywhere in the planes]**

Numba on Tycho's Zen 4 nodes.—Compiled for the AMD EPYC 9554 with every fastmath flag, numba's division-by-zero check in the Earth's Yukawa evaluation fired spuriously (the division itself was correct), failing the first potential-plane jobs. The fastmath kernels now use numpy's error model (`src/dp_core.py`); the solver's own finiteness checks remain.

Up-going only (resolved 2026-10-13).—The average now covers the whole sky with equal weights (Sec. 3.5); the per-direction output of Sec. 7 will add the detector's acceptance.

Implementation queue.—The core-count fix is done (2026-10-12): pools are sized from the process's CPU affinity capped by the SLURM allocation, not from the machine's core count. Tycho is set up (2026-10-13, `slurm/README.md`): an environment pinned to the laptop's versions, job scripts for the tests, the four-point validation (`tools/validate_record.py`) and the analysis; the test suite passes there. Next, in order: the per-direction solver output (the whole-sky average is done), the muon channel, the starting-event channel with the systematic uncertainties, then the statistical tests of Sec. 7.3. The details are in `docs/dev/WORK_PLAN.md`.

A Corrections

A.1 The relic-neutrino temperature (2026-10-12)

The default $T_{\nu,0}$ was 1.95×10^{-4} eV, which is 2.26 K: the value "1.95 K" had been entered as if it were in eV. The correct value is 1.676389×10^{-4} eV, so the relic neutrinos were 16% too hot (commit `5ecf10e`). We measured the effect on the absorption rates directly. Above every resonance, where the relics act as if at rest, the old temperature raised the rates by a factor of 1.553–1.570 across the energy range at the "below" point, close to $(1.95/1.676389)^3 = 1.574$. Across a resonant dip it raised the energy-integrated absorption by 1.573, the depth of the dip by 1.35–1.38 (against $(1.95/1.676389)^2 = 1.353$) and its width by 5–16%. Over the energies and redshifts of the "inband" grid, where the shapes of the dips also depend on the temperature, the ratio ranged from 1.33 to 5.4.

Effect on the fluxes.—At the benchmark points the fluxes changed by up to a factor of ~ 130 at individual energies. At "below" and "cosmoX2", where the flux is strongly absorbed

across the energy range, the median change was a factor of ~ 30, because a 57% change of the optical depth τ changes $e^{-\tau}$ by large factors once τ exceeds a few. Absorption features above the resonances moved down in energy, by 0.195 decades at "below" and "cosmoX2" (the relic-density factor, because the optical depth there scales as density over energy) and by ~ 0.15 decades at "inband". The resonant dips stayed at their energies ($E_{\rm res}$ does not depend on the temperature) but became shallower: the flux at the deepest "inband" dip rose by a factor of 3.6, at the deepest "strong" dip by 36%. The energy-averaged flavor fractions moved by at most 3.8×10^{-4} (below), 1.8×10^{-4} (inband), 7×10^{-9} (strong) and 6.7×10^{-5} (cosmoX2). Therefore, results of the flavor-ratio analysis obtained before the correction are almost unchanged, while spectra and event rates change substantially.

A.2 The cosmological and Galactic neutron densities (2026-10-12)

Cosmological.—The default $n_{n,0}$ was 2.5×10^{-7} cm^{-3}, documented as the neutron density but equal to the total baryon density (2.515×10^{-7} cm^{-3}, PDG 2024). The potential (T9) is sourced by neutrons alone: the vector couplings of the Z to protons ($\frac{1}{2} - 2\sin^2\theta_W$) and to electrons ($-\frac{1}{2} + 2\sin^2\theta_W$) cancel in neutral matter. The mediator Z_D inherits them through the mass mixing. Neutrons sit almost entirely in helium-4, which holds a fraction $Y_p = 0.2458 \pm 0.0013$ of the baryons (LBT Y_p project, 2026), half of them neutrons; deuterium adds $\sim 2 \times 10^{-5}$ per baryon. Therefore, $n_{n,0} = n_b Y_p/2 = 3.091 \times 10^{-8}$ cm^{-3}, and the cosmological potential was 8.1 times too strong.

Galactic.—The Galactic model converted the stellar mass density at half a neutron per nucleon (the factor $20.235 = 0.5 \times 40.464$ nucleons cm^{-3} per $M_\odot$ pc^{-3}) and the hot-halo density of Miller and Bregman (2013), which is an electron density, at 0.5 neutrons per electron. We now use 0.20 neutrons per nucleon for stars. Locally, stars and remnants amount to $33.4\,M_\odot$ pc^{-2}, of which 81% are visible stars, 15% white dwarfs and 4% brown dwarfs (McKee, Parravano and Hollenbach 2015). With $(Y + Z)/2 = 0.146$ for disc stars ($Y \approx 0.275$, $Z \approx 0.017$), 0.5 for carbon–oxygen white dwarfs and 0.142 for brown dwarfs, the weighted fraction is 0.20. For the hot gas ($Y \approx 0.26$, $Z \approx 0.3\,Z_\odot$, fully ionized) we use $(Y + Z)/2$ neutrons per $1 - (Y + Z)/2$ electrons, i.e., 0.15. The Galactic potential was thus 2.5 times (stars) and 3.3 times (gas) too strong.

Effect.—At the four benchmark points neither change alters the results beyond their floors: "below", "inband" and "strong" agree to $\leq 4 \times 10^{-11}$ in flux and $\leq 10^{-13}$ in flavor fractions; "cosmoX2" moves by 1.4×10^{-3} pointwise and 1.0×10^{-4} in the energy-averaged fractions, its known sensitivity to the last digit. At mediator masses of order MeV the coherent potentials are $\sim 10^{-22}$ of the vacuum term, so only the resonant absorption acts. (Correction, 2026-10-13: this holds for the cosmological and Galactic potentials changed here. The Earth's own potential, sourced by densities 24 orders of magnitude higher, is $\sim 10^{-8}$ eV at the benchmark couplings, far above the vacuum term; see Sec. 5.) The corrections matter for light mediators, where the potential grows as $1/m_{Z_D}^2$; we compared the old and new code at three such masses, with the default couplings. At $m_{Z_D} = 10^{-2}$ eV the total flux changes by up to 32%, the flavor fractions at single energies by up to 0.074 and the energy-averaged fractions by up to 0.013 (f_τ: $0.315 \to 0.328$). At 10^{-5} eV the changes are 69%, 0.144 and 0.004. At 10^{-8} eV, where the potential is so large that the flavor evolution is fully averaged, they are 3%, 0.008 and 10^{-4}.

A.3 Neutron fractions along Earth chords (2026-10-12)

Along Earth chords the code counted half of the nucleons as neutrons in every PREM layer (3.011×10^{23} g^{-1}). For neutral matter the neutrons per nucleon are $1 - Y_e$, where Y_e is the electron (equal to proton) fraction. We now use $Y_e = 0.4656$ in the iron–nickel core and 0.4957 in the silicate mantle and crust, the values used with PREM in neutrino-oscillation tomography, and the composition of water (8/18.015 neutrons per molecular mass) for PREM's top layer.

The core was thus 6.9% low, the mantle 0.9% low and the water layer 12.6% high. The regional boundaries coincide with PREM layer boundaries, so the phase-exact Earth steps stay exact (test `test_earth_neutrons_per_gram_by_layer`).

Effect.—The four benchmark points move: the energy-averaged ν_e and ν_μ fractions change by 0.025 (below), 0.026 (inband), 0.009 (strong) and 0.032 (cosmoX2), with the ν_τ fraction unchanged, and the flux at single energies by up to 21%. A change of a few per cent in the potential shifts the chord phases by $\sim 10^6$ rad, which redraws the residual conversion left by the 12-chord average. Adding one chord to the old code moves the same fractions by comparable amounts (Sec. 8). At light mediators the changes are smaller: up to 3.8×10^{-3} in the energy-averaged fractions at $m_{Z_D} = 10^{-2}$ eV and below 10^{-5} at 10^{-5} and 10^{-8} eV.

A.4 Point sources and cold disc gas (2026-10-12)

The Sun, the Moon and the Earth enter the Milky Way and Earth phases as point sources of a Yukawa potential, which reaches the detector only when the mediator's range, $1/\sqrt{m_{Z_D}^2 + m_D^2}$, spans the distance to the body. Their neutron numbers were inconsistent: 0.248 neutrons per nucleon for both the Sun and the Moon, 0.496 for the Earth. The Sun's present bulk composition is the protosolar one (X, Y, $Z = 0.7154$, 0.2703, 0.0142; Asplund et al. 2009) plus the $\sim 0.038\,M_\odot$ of hydrogen it has burned into helium, which we estimate from its luminosity integrated over 4.57 Gyr; that gives $Y = 0.308$ and $(Y + Z)/2 = 0.161$ neutrons per nucleon, 1.929×10^{56} in all. The Moon is silicate rock with a small iron core, 0.505 per nucleon, 2.232×10^{49}. For the Earth we integrate PREM with the per-layer fractions of the chords (Appendix A.3); PREM's mass comes out at 5.973×10^{27} g, within 0.02% of the Earth's, with 1.849×10^{51} neutrons (0.514 per nucleon).

The Galactic model had no cold gas. We add the HI and H_2 discs of McMillan (2017, arXiv:1608.00971), $\rho = \Sigma_0/(4z_d)\, e^{-R_m/R - R/R_d}\, \mathrm{sech}^2(z/2z_d)$ with $(\Sigma_0, R_d, R_m, z_d) = (53.1\,M_\odot\,\mathrm{pc}^{-2}, 7, 4, 0.085$ kpc) and $(2180\,M_\odot\,\mathrm{pc}^{-2}, 1.5, 12, 0.045$ kpc). Their surface densities already include helium and metals; at the Sun they give 10 and $2\,M_\odot$ pc^{-2}, against the $13.7\,M_\odot$ pc^{-2} that McKee et al. (2015) measure in all gas. We convert them at $(Y+Z)/2 = 0.142$ neutrons per nucleon (protosolar composition).

Effect.—At the four benchmark points nothing changes beyond the floors ($\leq 4 \times 10^{-11}$ in flux; "cosmoX2" 9×10^{-4} pointwise), because at MeV masses the point sources are out of range and the Galactic potential is negligible. At light mediators the changes appear at individual energies: at $m_{Z_D} = 10^{-2}$ eV up to 18% in flux and 0.064 in the flavor fractions, at 10^{-8} eV up to 3% and 0.008. We also ran $m_{Z_D} = 10^{-20}$ eV, where the bodies are within range: with Debye screening off the flux changes by up to 27% and the flavor fractions by up to 0.13; with $g_D = 10^{-12}$, by 1.6% and 0.004. The energy-averaged fractions move by at most 1.2×10^{-3} in all these runs.

A.5 The length of the Earth chords (2026-10-12)

A chord from the detector, at radius $R - d$, meets the surface after a length u that solves $u^2 + 2(R-d)\cos\theta_z\, u - d(2R-d) = 0$. The function `get_baseline_earth` dropped the terms in d^2, so for $d = 1.5$ km every chord but the vertical one started above the ground: by 8.1 m for the horizontal chord, 1.9 m at $\cos\theta_z = -0.09$ and less than a metre for steeper ones. For 10 of the 12 default chords that vacuum stretch was shorter than the margin with which the layer-crossing search discards the entry point (10^{-6} of the chord), so the step integration extended the top layer to the start of the chord. The phase-exact scheme remained step-count independent, since the same stretch was added for every step count, but it integrated up to 0.93 m too much water, a phase of up to 0.4 rad at "below". We now solve the quadratic exactly, in a form without cancellation for either sign of $\cos\theta_z$; every chord starts on the surface to 3×10^{-12} km (test `test_exact_chord_baseline_starts_on_the_surface`).

Effect.—The energy-averaged flavor fractions change by 9.1×10^{-4} (below), 4.5×10^{-5} (inband) and 1.5×10^{-5} (strong), the flux at single energies by at most 8.7×10^{-3}, 4.1×10^{-4} and 4.5×10^{-5}, with the dips unchanged. "cosmoX2" moves by 4.1×10^{-3} in the averaged fractions, a redraw of its phase noise. Among the light mediators only $m_{Z_D} = 10^{-2}$ eV, also phase-noise dominated, moves visibly (8.5×10^{-3}); the ultralight points do not change.

A.6 The Earth's own potential (2026-10-12)

Until 2026-10-12 the Earth chords used the local density, the limit of Eq. (1) when $1/m$ is far below every length over which PREM varies. A mediator with a range of 200 m ($m \approx 10^{-9}$ eV) already sees the layer boundaries smoothed; once the range rivals the Earth, the potential is set by the enclosed neutrons and the local form overstates it by $\sim 1/(mR)^2$, 10^{13} at $m = 10^{-20}$ eV.

Evaluation.—Writing $f = rn$, Eq. (1) becomes $n_{\rm eff} = (m/r)\,[P + (1 - e^{-2mr})Q/2]$, with $P(r) = \int_0^r f(r')\,\frac{1}{2}(1 - e^{-2mr'})\,e^{-m(r-r')}dr'$ and $Q(r) = \int_r^R f(r')\,e^{-m(r'-r)}dr'$, where every exponential decays; e^{mR} itself would overflow at any MeV mass. For $mR < 50$ we accumulate P outward and Q inward on subintervals no longer than $1/m$, with 10-point Gauss–Legendre quadrature, which is exact for PREM's polynomials to rounding. For $mR \geq 50$ we integrate by parts, again exactly for polynomials: inside a shell $(a, b]$, $n_{\rm eff} = n + (f''/m^2 + f''''/m^4)/r + (m/r)[\delta\, e^{-m(r-a)} + \gamma\,(1 - e^{-2mr})\,e^{-m(b-r)}/2]$, with one pair of constants δ, γ per shell. The last term carries the boundary layers; away from them every exponential underflows and $n_{\rm eff}$ equals n to $(mR)^{-2}$. At the benchmark masses the result is the local density bit for bit.

Tests.—The suite checks the two evaluations against the closed form for a uniform ball from $mR = 10^{-7}$ to 10^8 (relative 10^{-12} for the tabulated one, 10^{-13} of the density for the other), against direct quadrature of the shell kernel for PREM (ranges from $R_\oplus$ to 200 m, 10^{-9}), against each other across the switch, and against the enclosed-charge limit for $mR = 10^{-12}$. Outside, the integral is a point source of $4\pi \int r^2 n\, \sinh(mr)/(mr)\, dr$ neutrons, which equals the Earth's 1.849×10^{51} to 10^{-3} where the Milky Way phase can reach it. Along the chords, the step-summed integral is the same for 8, 25 and 1000 steps to 10^{-12}, with ranges from 200 m to 2×10^6 km. The boundary layers carry a floor of their own. Near a layer the radius is known only to $\sim 10^{-12}$ km, so a layer of thickness $1/m$ is integrated to a relative accuracy of $\sim m \times 10^{-12}$ km (5×10^{-6} at 10^{-3} eV), an error below 10^{-12} of the step it lies in. Layers thinner than 10^{-10} of a step are left unresolved, an error below 10^{-10} of that step's phase.

Effect.—At the four benchmark points the fluxes are identical, bit for bit, to those of the chord-length fix alone, as are the step densities of all 12 chords. At light mediators, $m_{Z_D} = 10^{-2}$ eV moves by 2.7×10^{-3} in the averaged fractions (phase noise), 10^{-5} eV by 1.6×10^{-6} and 10^{-8} eV by 5×10^{-6} (flux at single energies up to 2%). The ultralight points do not change: there the Galactic potential is already so large that the state reaching the Earth is fully averaged in the basis of the potential, whose texture the Earth's potential shares. To see the effect where the Earth matters, we ran three points with ranges of $R_\oplus$, 100 km and 1 km at fixed $g_D\epsilon'/m^2$ ($m_{Z_D} = 3.1 \times 10^{-14}$, 1.97×10^{-12} and 1.97×10^{-10} eV; $g_D\epsilon' = 10^{-49}$, 4×10^{-46}, 4×10^{-42}). In the local form all three accumulate 52 rad along the vertical chord and 1.6 rad at $\cos\theta_z = -0.09$; the Yukawa integral reduces these to 8.7 and 0.4 rad at the range $R_\oplus$, to 52.0 and 1.0 rad at 100 km (the grazing chords run through the thin top layers), and leaves them at 1 km. With no potential the averaged fractions are $(0.3037, 0.3543, 0.3420)$; the local form shifts f_e by $+0.0224$ at all three, because it depends only on $g_D\epsilon'/m^2$. The Yukawa integral gives $+0.0240$, $+0.0207$ and $+0.0224$: the Earth's effect changes by 7–8% once the range reaches 100 km. At these points 25 and 200 steps agree to 1.8×10^{-10} in flux and 2.6×10^{-11} in the fractions.

A.7 The zenith average of the Earth chords (2026-10-12)

The flux at the detector is the average over the up-going hemisphere, uniform in $\cos\theta_z$, which we take from a finite set of chords. Wherever the dark-portal phase across the Earth matters, the potential dominates the vacuum term: a phase of 1 rad over the Earth needs $V \sim 10^{-14}$ eV, against $\Delta m^2/2E \lesssim 10^{-16}$ eV above 10 TeV. Therefore a chord evolves as $U(c) = \sum_g e^{-i\lambda_g \Psi(c)} P_g$, with P_g the eigenprojectors of the texture (eigenvalues λ_g) and $\Psi(c) = \mathrm{Coef} \int n\, dl$ the chord's column of potential, and each block of the evolved density matrix is $P_g \rho P_h = A_{gh}(c)\, e^{-i\omega_{gh}\Psi(c)}$, $\omega_{gh} = \lambda_g - \lambda_h$, with an amplitude A_{gh} that varies slowly with c (the vacuum term's own phases across the Earth are below 10^{-2} rad). At MeV masses Ψ reaches 10^5–10^8 rad, so the cross blocks have effectively random phases from chord to chord, and the equal-weight mean we used kept them as sampling noise. Adding a thirteenth chord to twelve moved the averaged fractions by up to 0.048 (commit `0db8111`), $\sim 10^{-2}$ at the analysis's 50 chords, which exceeds the projected 2050 width of f_μ (≈ 0.005). The phase does not depend on energy, so the noise did not average out over the spectrum.

Method.—We now integrate each block against its known phase. A_{gh} is interpolated linearly between the chords, and the oscillatory factor is integrated exactly on a fine grid of $\Psi(c)$ with the phase linear on each subinterval (a Filon-type rule). The result is a complex weight per chord and block, $w_k^{gh} = \int \mathrm{hat}_k(c)\, e^{-i\omega_{gh}[\Psi(c)-\Psi(c_k)]}\, dc$, the same for every energy. The fine grid has 4096 chords, with points added toward the cosines at which a chord starts to cross a PREM boundary, where Ψ' diverges; their columns take ~ 10 ms once per process. For $\omega = 0$ the weights are the trapezoid rule. Where the phase turns many times between neighbouring chords they fall to $\sim 1/(\omega\Psi')$, and the average becomes the decoherent $\sum_g P_g \rho P_g$ that the continuum of zeniths produces. In between they resolve the oscillation that a chord sum aliases. The rule needs the phase-exact steps: under the historical left-endpoint scheme, which keeps the equal-weight mean, each chord's phase carries a Riemann error of thousands of radians.

Tests.—For chords evolved exactly as $e^{-i\Psi S}$, the average of 12 or 13 chords matches the integral over $\cos\theta_z$ to $\lesssim 10^{-5}$ for Earth phases from 0.1 to 10^8 rad (a dense trapezoid over 200,001 chords up to 10^3 rad, the block-diagonal limit at 10^8 rad; at 10^5 rad the continuum keeps a cross term of 3×10^{-4}, which the weights reproduce to 10^{-5}). The equal-weight mean misses by 10^{-3}–10^{-1}. The suite also checks the trapezoid limit, Hermiticity for a full-rank complex texture, and the options.

Effect.—At production settings, 12, 13 and 25 chords agree with 50 to 6×10^{-9} (below), 1.4×10^{-7} (inband) and 5×10^{-8} (strong) in the energy-averaged fractions; with the equal-weight mean, 12 and 13 chords differed by 4.1×10^{-3}, 1.2×10^{-3} and 5.6×10^{-4}. The step-count certificate tightens to $\leq 1.2 \times 10^{-10}$ in flux. Against the equal-weight mean of 12 chords, the fractions move by 0.029 (below), 0.018 (inband) and 0.0057 (strong). At "below" they become $(0.3192, 0.3192, 0.3617)$, against $(0.2904, 0.3480, 0.3617)$: with $u_e = u_\mu$ the random Earth phases mix ν_e and ν_μ equally, and the asymmetry was noise. At $m_{Z_D} = 10^{-2}$ eV the chord count no longer matters at all (10^{-16}), and in the coherent window of Appendix A.6 the fractions move by 10^{-3}–4.5×10^{-3} and agree across chord counts to 5×10^{-9}. "cosmoX2" keeps a residual of $\sim 6 \times 10^{-4}$, down from 5×10^{-3}, for the reason given in Sec. 8. A solver run costs the same as before. (The "cosmoX2" residual was the propagator's non-unitarity and is gone since commit `2066f3b`, Appendix A.10.)

A.8 The cosmological cascade (2026-10-12)

Each absorption at an energy E_j re-emits a neutrino at a lower energy E with the spectrum $3E^2/E_j^3$. Within a redshift step the code held this regenerated source at its value at the start of the step, ΓF_n. Near a resonance the optical depth of one step, $\Gamma\Delta t$, reaches 10^{10}, and it exceeds 1 in 70% of the (z, E) cells at "inband". A bin then re-emitted up to $\Gamma\Delta t$ times more neutrinos than it absorbed. The excess compounded from step to step: at "inband" the energy reaching

the Earth was 5×10^{36}, 8×10^{60} and 6×10^{66} times the no-interaction value with 100, 1000 and 10,000 redshift steps (at "strong", 6×10^{9} to 10^{15}). The midpoint weights of the kernel also re-emitted $1 + 1.5\,\Delta \ln E$ neutrinos per absorption, 1.2% too many per generation at 1000 bins. The flavor fractions, dominated by the runaway secondaries, stayed near 1/3 and looked plausible. Every earlier validation compared the code with a previous version of itself, which shared the error; the first check against conservation found it.

Fix.—We now sweep the energy bins of each step from high to low. A bin solves $dF/dt = -\Gamma F + S$ exactly over the step, with S the injection plus the regeneration from higher bins, computed from the number those bins actually absorbed during the step, $F_n(1-e^{-\Gamma\Delta t}) + S\,[\Delta t - (1 - e^{-\Gamma\Delta t})/\Gamma]$. A bin's regeneration into itself is solved implicitly. The kernel is integrated exactly over the bins (commit `7dc95e9`). A step conserves the number of neutrinos to rounding for $\Gamma\Delta t$ from 10^{-8} to 10^{10}, and the solver never returns more energy than it injects; tests check both.

Effect.—With the absorption of the time, the energy reaching the Earth fell to 3.0×10^{-3} (below), 3.3×10^{-3} (inband), 0.11 (strong) and 3.0×10^{-3} (cosmoX2) of the no-interaction value. Absorption that strong at couplings of 10^{-4} is implausible; we traced it to the normalization of Appendix A.9.

A.9 The normalization of the resonant absorption (2026-10-12)

The rates integrate a Breit–Wigner cross section over the relic distribution. The theory notes give its width, $\Gamma = (\Sigma G)M/(12\pi)$ (T11), and the couplings ΣG (T12), but not its numerator. The code used $\sigma_\alpha = 2\pi(\Sigma G)_\alpha s/[(s - M^2)^2 + M^2\Gamma^2]$, with a single coupling factor. Its integral over s, $12\pi^2(\Sigma G)_\alpha/\Sigma G$, does not depend on the size of the couplings: over the benchmark grid the integrated rate was 1.870×10^{-17} eV whether ΣG was 7.6×10^{-4} or 3×10^{-19}. Any nonzero mixing, however small, absorbed as strongly as $u = 0.5$; only $u = 0$ exactly recovered standard propagation.

Fix.—We decided (2026-10-12) on the Breit–Wigner cross section of a vector resonance formed by $\nu_L + \bar{\nu}_R$, summed over final states,

$$\sigma_\alpha = \frac{48\pi\,\Gamma_{\mathrm{in},\alpha}\,\Gamma\,s}{M^2[(s - M^2)^2 + M^2\Gamma^2]} = \frac{(\Sigma G)_\alpha(\Sigma G)\,s}{3\pi[(s - M^2)^2 + M^2\Gamma^2]}, \qquad \Gamma_{\mathrm{in},\alpha} = \frac{(\Sigma G)_\alpha M}{12\pi}, \tag{3}$$

whose integral over s is $4\pi(\Sigma G)_\alpha$; for the Z it gives the classic $\langle\sigma\rangle = 2\sqrt{2}\pi G_F$. The rates are the old ones times $\Sigma G/(6\pi^2)$, applied where the grids leave `src/dp_thermal.py`, so the coupling cache keeps working on the unnormalized grids (commit `944360d`). We test that the integrated rate is proportional to ΣG at fixed flavor ratios and vanishes at $u = 0$, the narrow-width limit against an independent quadrature (to 2×10^{-3}), and that $u = 10^{-10}$ reproduces the $u = 0$ flux to 10^{-8}.

Effect.—With the cascade fixed, the energy reaching the Earth is 0.999 (below), 0.841 (inband) and 0.982 (strong) of the no-interaction value (Sec. 4). In July we had rejected rescaling the rate grid across coupling scans because the rates at the resonance did not scale with the couplings; that independence was this error.

A.10 The unitarity of the spatial propagator (2026-10-12)

The propagator evaluated $U = e^{-iH\Delta x}$ as a polynomial in the traceless part of H, through Newton divided differences of $e^{-i\lambda\Delta x}$ at the computed eigenvalues. A polynomial evaluated at rounded eigenvalues is unitary only to $\sim \varepsilon\,\Delta\lambda\,\min(\Delta x, 2/\delta\lambda)$, with $\Delta\lambda$ the spread and $\delta\lambda$ the smallest gap of the eigenvalues. Under the potential of "cosmoX2", ~ 70 eV in the Earth's core, the error reached 5×10^{-3} per Earth step: the Milky Way and Earth transits changed the total flux of an energy bin by -0.5% to $+0.7\%$. The energy reaching the Earth exceeded the no-interaction value (1.0003), which absorption alone cannot do. A guard added in July

for $m_{Z_D} \sim 10^{-6}$ eV, where the error grew to $\sim 10^{11}$ and overflowed to NaN, switched to an exactly unitary form only once the error exceeded 0.1. The polynomial also read both triangles of an H that is Hermitian only to rounding. For single-flavor mixing ($u_\mu = 0$ or $u_e = 0$) above $\sim 10^7$ eV, its basis for nearly degenerate pairs divided by zero.

Fix.—We now evaluate U as the spectral sum over an eigenbasis that stays orthonormal to rounding at any gap (Sec. 3; commit `2066f3b`). It is unitary to 2×10^{-15} over 12,880 production textures, with potentials from 10^{-24} to 10^{22} eV and steps up to 6×10^{32} eV^{-1}, and over 20,000 random spectra; the polynomial reached errors of order one in the same sweep. Against a 50-digit matrix exponential we find it as accurate as the polynomial where the latter was sound. A pair whose splitting is below $16\varepsilon(|q|+p)$, with q and p the mean and the scale of the eigenvalues, evolves as degenerate. Rounding when H is formed already splits an exact degeneracy by up to $3.1\varepsilon(|q|+p)$; evolving that splitting would rotate the pair by tens of radians per Earth step. The new form costs 0.6–0.7 times the old one per Earth step and 1.25 times per Milky Way step.

A stale compilation.—Our first check of the fix returned the old totals and the old division by zero. numba validates a cached function against its own source file only, so the spatial kernels, which call the propagator from another file, had kept its old machine code. We now key the cache by a digest of every source file (`src/dp_core.py`), so any edit recompiles everything.

Effect.—The spatial phase conserves every total to 10^{-14}. At "below", "inband" and "strong" the results are unchanged to five digits. At "cosmoX2" the energy reaching the Earth falls from 1.0003 to 0.99934 of the no-interaction value. The fractions move from $(0.32917, 0.32917, 0.34166)$ to $(0.32899, 0.32899, 0.34201)$. The point no longer depends on the last digit, on the number of chords or on the number of Earth steps (Sec. 4 and 8).

A.11 Both secondaries of every absorption (2026-10-13)

An absorption $\nu(E_j) + \bar{\nu}_{\rm relic} \to Z_D \to \nu + \bar{\nu}$ emits two particles. The Z_D is polarized along the incoming neutrino, so the outgoing neutrino carries the spectrum $3E^2/E_j^3$ (on average $\frac{3}{4}$ of E_j) and the antineutrino $3(E_j - E)^2/E_j^3$ (the remaining $\frac{1}{4}$); an absorbed antineutrino gives the mirror image. The code kept only the secondary of the absorbed particle's own kind and propagated neutrinos and antineutrinos independently, as Eq. (T6) of the theory notes does, which regenerates ρ only from ρ. Half of the regenerated particles and a quarter of the absorbed energy were dropped.

Fix.—We now evolve neutrinos and antineutrinos together through the cosmological phase. Within a redshift step, each energy bin receives both kinds of secondaries from the absorptions above it; the second kernel, integrated exactly over the bins, separates into three running sums over the parents, and a bin's own secondaries of both kinds make each bin a 2×2 linear system, which we solve directly. The Milky Way and Earth transits stay separate for the two kinds. A step balances the number of particles to rounding for $\Gamma\Delta t$ from 10^{-8} to 10^{10}: every absorption adds one particle, less the secondaries that fall below the grid (a test). The analysis evolves the configured sources in one coupled run, and the responses to neutrino and antineutrino sources separately only when their ratio floats.

Effect.—At production settings the energy reaching the Earth rises from 0.841 to 0.941 of the no-interaction value at "inband" and from 0.982 to 0.99995 at "strong", and the number from 0.955 to 1.092 and from 0.99993 to 1.00074: what is left is the energy of secondaries below 10 TeV. The energy-averaged fractions move by 1.8×10^{-3} at "inband" (f_τ: $0.33801 \to 0.33621$) and by $\leq 2 \times 10^{-5}$ at the other points.

On the theory notes.—Section 4 of the theory notes asks for a factor $\frac{1}{2}$ in the absorption rate for identical initial-state background particles. We do not apply it, and we think the sentence should be reworded: the rate of one high-energy neutrino counts each relic it can annihilate with once (one helicity state, 56 cm^{-3} per mass state), which is Eq. (T7) as written; a factor $\frac{1}{2}$ belongs

to final states integrated over all angles, where it is part of the width (T11), and to collision rates within one gas of identical particles. Equation (T6) should include the regeneration of ρ from $\bar{\rho}$ described above.

A.12 The transit to the Earth (2026-10-13)

The potential planes of Sec. 5 scan the coupling product $P = g_D\epsilon'|U|^2$ down to ultralight mediators. Their lower-left corner scattered: at $(m_{Z_D}, P) = (10^{-27}\ \text{eV}, 10^{-28})$ the χ^2 was 1.83 with the default six Milky Way directions, 14.49 with 50 and 15.07 with 200. We traced the scatter, and the physics around it, to four errors in the transit from the cosmological arrival to the Earth, and to a fifth in the cosmological potential.

Precision.—At large couplings the potential exceeds the vacuum term by up to 10^{27}. The rank-one texture $uu^\dagger$ has two zero eigenvalues, split only by the vacuum term, and any method that forms the Hamiltonian first (the closed-form eigenvectors, LAPACK) computes its eigenvalues with an absolute error $\varepsilon\sigma$: above $\sigma/M \sim 10^{13}$ the two states orthogonal to u were rounding noise. The noise entered the cosmological mixing table and the start of every Milky Way path, and the average over directions converged on the average of the noise. We now diagonalize by Jacobi rotations in the texture's eigenbasis, where the large entries sit on the diagonal and the degenerate eigenvalues are exactly zero (`eigh_graded`): eigenvalues and eigenvectors agree with 60-digit arithmetic to 3×10^{-15} up to $\sigma/M = 10^{40}$, where LAPACK is wrong already at 10^{16} (a test, for rank-one, full-rank and degenerate textures).

The Galactic potential.—The Milky Way phase used the local density at every mediator mass. That is the Yukawa integral only for ranges short against every length over which the density varies; for longer ranges the potential at the Sun is set by the neutrons within the range, and the local form overstated it by 5×10^6 at 10^{-30} eV, 750 at 10^{-28} eV, 19 at 10^{-27} eV and 3 at 10^{-26} eV. We now evaluate $n_{\text{eff}}(m) = \int_0^\infty m^2 r\, e^{-mr}\, \bar{n}(r)\, dr$, with $\bar{n}(r)$ the model's density averaged over the sky at distance r from the Sun, tabulated once by composite Gauss–Legendre quadrature graded toward the plane and the Galactic Centre (accurate to 4×10^{-9} against a finer rule; `tools/mw_sky_density.py`), plus the cosmic mean beyond the Galaxy, where the halo falls below it (336 kpc). The model's exponential discs have a cusp at $z = 0$, where the Sun sits, so n_{eff} approaches the local density only as $1/m$; above 10^{-12} eV it is the local density itself, continuous to 1.5×10^{-14}.

Adiabaticity.—The Galactic potential changes over ≥ 10 pc, while the oscillation length is ≤ 0.005 pc even at 30 PeV, so the transit is adiabatic: each eigenstate keeps its population. The 25 left-endpoint steps of 4 kpc turned every step boundary into a sudden jump of the eigenbasis, an error that fell only as the inverse step count: at $(10^{-12}\ \text{eV}, 10^{-24})$ the χ^2 was 6.6, 10.6 and 13.4 with 25, 200 and 1000 steps. We now map the arriving populations onto the local eigenstates directly, which is exact in the adiabatic limit and independent of the direction. At "cosmoX2" ($m_{Z_D} = 40$ eV, where the Galactic potential is up to 0.7% of the solar vacuum term) the old method needed 50 directions to converge onto the new result (4×10^{-6}); at its defaults it was 1.4×10^{-3} off pointwise.

The Sun and the Earth's exterior.—Per unit P at the Earth, the Sun's potential exceeds the Galaxy's above $\sim 10^{-26}$ eV, as long as its range spans an AU ($\lesssim 10^{-18}$ eV), and the Earth's own potential dominates above that. The Milky Way phase sampled the Sun only at its 4-kpc steps; the Earth phase then switched it on as a constant along the chords, and the chords started at the Earth's surface, without its exterior. Both potentials turn on over distances comparable to the oscillation lengths (0.07–6600 AU between 10 TeV and 30 PeV; the Earth's $1/r$ tail reaches hundreds of Earth radii before it falls below the vacuum term near the planes' boundary), so their transit can be adiabatic, sudden or in between. We integrate it in the instantaneous eigenbasis. The eigenvectors of $(M+\sigma S)/2E$ depend on σ only, so one table serves every energy and path, with nodes no more than $\Delta\theta = 0.002$ of eigenbasis rotation apart; across a step the dynamic phases are integrated exactly and the non-adiabatic coupling K/h, $K = \log(W_n^\dagger W_{n+1})$,

against the phase difference (first-order Magnus with a Filon rule, exact in the adiabatic and sudden limits). Against a fourth-order Magnus integration that resolves every oscillation (up to 7×10^6 steps) through a $1/r$ potential that is neither adiabatic nor sudden, the results agree to $\sim 10^{-4}$ at $\Delta\theta = 0.01$ and $\sim 10^{-5}$ at 0.002, falling linearly with $\Delta\theta$, in 0.2–1 ms per path instead of seconds (tests). The Sun's transit is averaged over eight Gauss–Legendre cosines of the angle to it; the Earth's exterior beyond $2R_\oplus$ is one radial path for all chords, inside it each chord's own line in eight steps, and each chord adds its line's extra column beyond $2R_\oplus$ as a phase. The Moon, at most 2×10^{-4} of the Earth's surface potential, stays a constant background.

The cosmological potential.—`calc_V_LRI_cosmo` and the mixing table used $1/\max(m_{Z_D}^2 + m_D^2, 10^{-25}\ \mathrm{eV}^2)$, a guard against division by zero added on 2026-07-20 with the FrequentistAnalysis adaptation, which capped the cosmological potential for every mediator lighter than 3×10^{-13} eV (10^{35} times too weak at 10^{-30} eV). The decided replacement is the uniform medium's Yukawa integral cut at the Hubble radius, $(1 - (1 + x)e^{-x})/m^2$ with $x = mR_H(z)$: $1/m^2$ above $\sim 10^{-32}$ eV, $R_H^2/2$ below, with no division by zero.

Convergence.—On the $e\mu$ potential plane, at points chosen where each source dominates near the boundary of the excluded region and at the old scattered points, χ^2 moves by at most 0.03 from 8 to 32 directions to the Sun and from $\Delta\theta = 0.002$ to 0.001, by at most 0.05 from 400×200 to 800×400 energy and redshift bins, and by at most 0.002 when the split radius doubles and the exterior steps quadruple. The old scattered points are now plateaus: $\chi^2 = 57.19$ at both $(10^{-27}\ \mathrm{eV}, 10^{-28})$ and $(10^{-25}\ \mathrm{eV}, 10^{-28})$, and 14.36 at $(10^{-12}\ \mathrm{eV}, 10^{-24})$, under every setting.

Effect.—At the benchmark points below, inband and strong the fluxes agree with the previous solver to 3×10^{-12}; at cosmoX2 they move by 1.4×10^{-3} pointwise and the fractions by 1.8×10^{-8}, the old method's own error (above). The potential planes are recomputed in Sec. 5.

What this does not model.—The cosmological phase uses the cosmic mean density everywhere, including at the sources, whose own environment would set the projection of the injected flavors for short ranges. The Local Group beyond the Galaxy (M31 adds $\sim 1\%$ for ranges beyond 1 Mpc) is not included. The Sun's and the Earth's averages are taken independently, as if the Sun's direction and the zenith angle at the South Pole were uncorrelated. Inside the Earth, under a potential 10^{14} times the vacuum term, the propagator still evolves the two states orthogonal to the potential as degenerate (Sec. 3), a phase error of at most 8×10^{-3} rad across the Earth at 10 TeV.

A.13 The mixing-parameter priors (found 2026-10-12)

The Gaussian priors in the `fallbacks` block of the analysis configurations have never been applied, for three reasons. First, the configurations name the shape `"gaussian"`, while the code accepts only `"gauss"`. Second, the entries `dcp` and `dm3l_sq` do not match the parameter names `d_cp_deg` and `dm31_sq`. Third, the prior centres disagree with the fixed values (δ_{CP} 212° against 132°, Δm_{21}^2 7.537×10^{-5} against 7.41×10^{-5} eV^2, which is 3.4σ of the prior). In addition, the branch for angles in degrees converts a value that is already in degrees. The NuFIT 6.1 constraints of Sec. 7.2 will replace these priors. FrequentistAnalysis was retired on 2026-10-12; the analysis package that replaced it applies its priors and checks their names against the model's parameters (Sec. 5).

A.14 Earlier corrections

Thermal kernel (July 2026).—The midpoint-rule momentum integral erred by 6–9% at on-resonance cells; validation against the previous code could not reveal this, since it measures only differences. A dense reference integration did. The semi-analytic kernel replaced it, which changed the fluxes at the dips by up to 15% with flavor ratios within 10^{-6} and dip energies unchanged (round 7).

Earth steps (July 2026).—The left-endpoint stepping scattered the results at the 10^{-2} level under any change of the step count (Sec. 3.5); in-band energy-averaged flavor ratios carried a 1.7×10^{-2} ambiguity at the analysis settings. The phase-exact scheme removed it.

Coupling cache (July 2026).—The cache never activated in real scans (Sec. 3.2); its narrower brackets fixed that without loss of accuracy.